



CM

On-Load Tap-changer

Operating Instructions

HM 0.460.301-02.02/2021



IMPORTANT:

*Please read this manual carefully
before using the tap-changer and
save it for reference.*

Thank you for choosing Huaming products.

Before using, please read this instruction carefully. The product may have been modified since this document was published. We reserve the right to change the technical data, design and scope of supply without warning in advance.

Generally, information provided with individual quotations and orders is binding. Huaming has no liability for any damage due to incorrect interpretation of this document.

The original document is written in Chinese.

Keep this manual at hand for future use and in case of any question, please contact Huaming.

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1.Foreword

This instruction manual is a comprehensive description of product transportation, installation, commissioning, test and maintenance. It is intended for trained and qualified staff related to transformer production, testing and quality assurance, and has reference value for transformer design and operation personnel.

1.1 Safety

Safety instructions are explained in this manual. They must be observed and followed to avoid relevant dangers.

Personal safety equipment must be worn during work to prevent potential risks.

	Protective clothing for protecting personnel from hazards in the work environment.
	Safety shoes for protecting personnel from falling sharp objects and slippery surfaces.
	Goggles for protecting eyes and facial area from impact, dust, liquid splash and high heat hazards.
	Hard hat for protecting head from injury.
	Hearing protection for protecting hearing from damage.
	Protective gloves for protecting hands from mechanical, chemical, and electrical hazards.

Table 1 Personal safety equipment list

1.2 Notation and conventions



DANGER!

The yellow triangle warning sign means life and health may be at risk if neglecting the instruction. Neglecting this warning can lead to serious or fatal injuries.



CAUTION!

The orange round warning symbol means products or device may be damaged if the instruction is not observed. Neglecting this warning can lead to product's or operator's damages and injuries.

**NOTE!**

The blue note symbol indicates important information that must be noted.

1.3 Safety precautions

Precaution	Action
Electric shock	Always unplug a device before inspection. Stay away from high-voltage equipment and wear protective clothing.
Transformer oil	Collect used transformer oil in dedicated drums. Keep it away from flames or sparks. Do not touch it which may cause damages to health.
Fire	In case of fire, extinguish it with powder, foam or carbon dioxide. Do not use water.
Waste	Wastes are dangerous to the environment or human health. Clean them up in time and use different methods according to the wastes' categories.

Table 2 Safety precaution

2. General

CM on-load tap-changer, used to regulate voltages, is suitable for power or industrial transformers with rated voltages of up to 500kV, a maximum rated current of three-phase 600A, single-phase 1500A, and a frequency of 50Hz or 60Hz. The three-phase tap-changer is used for wye-connection (star connection) at neutral point, and the single-phase for any voltage regulation.

CM tap-changer is a typical combined tap-changer with a diverter switch and a separate tap selector (Figure 1). The diverter switch is composed of a switch insert and an oil compartment with insert installed inside.

CM OLTC is installed on the transformer tank cover via the tap-changer head flange, and is connected to the CMA7 or SHM-D motor drive unit through the worm gear reducer, horizontal drive shaft, bevel gear box, and vertical drive shaft, so as to achieve the purpose of tap change.

**NOTE!**

Special design is available. Please contact Huaming for further information.

General view

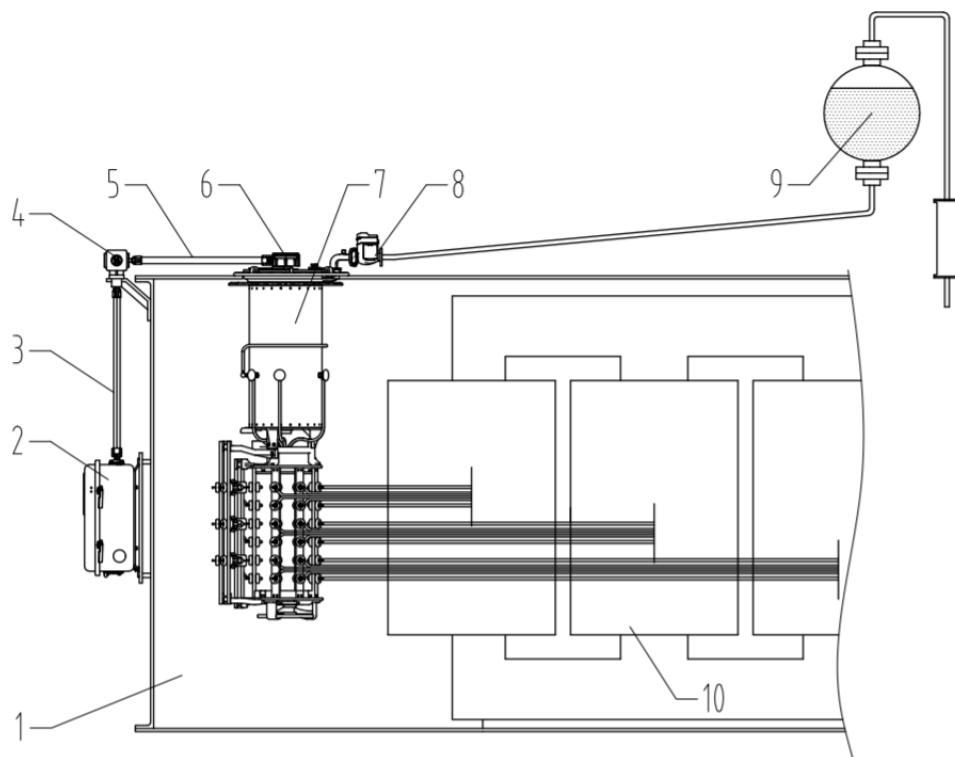


Figure 1 Overview of tap-changer inside transformer

1. Transformer tank	2. Motor drive unit
3. Vertical drive shaft	4. Bevel gear box
5. Horizontal drive shaft	6. Upper gear unit
7. Tap-changer	8. Protective relay
9. Oil conservator	10. Active part of the transformer

Table 3 List of all parts

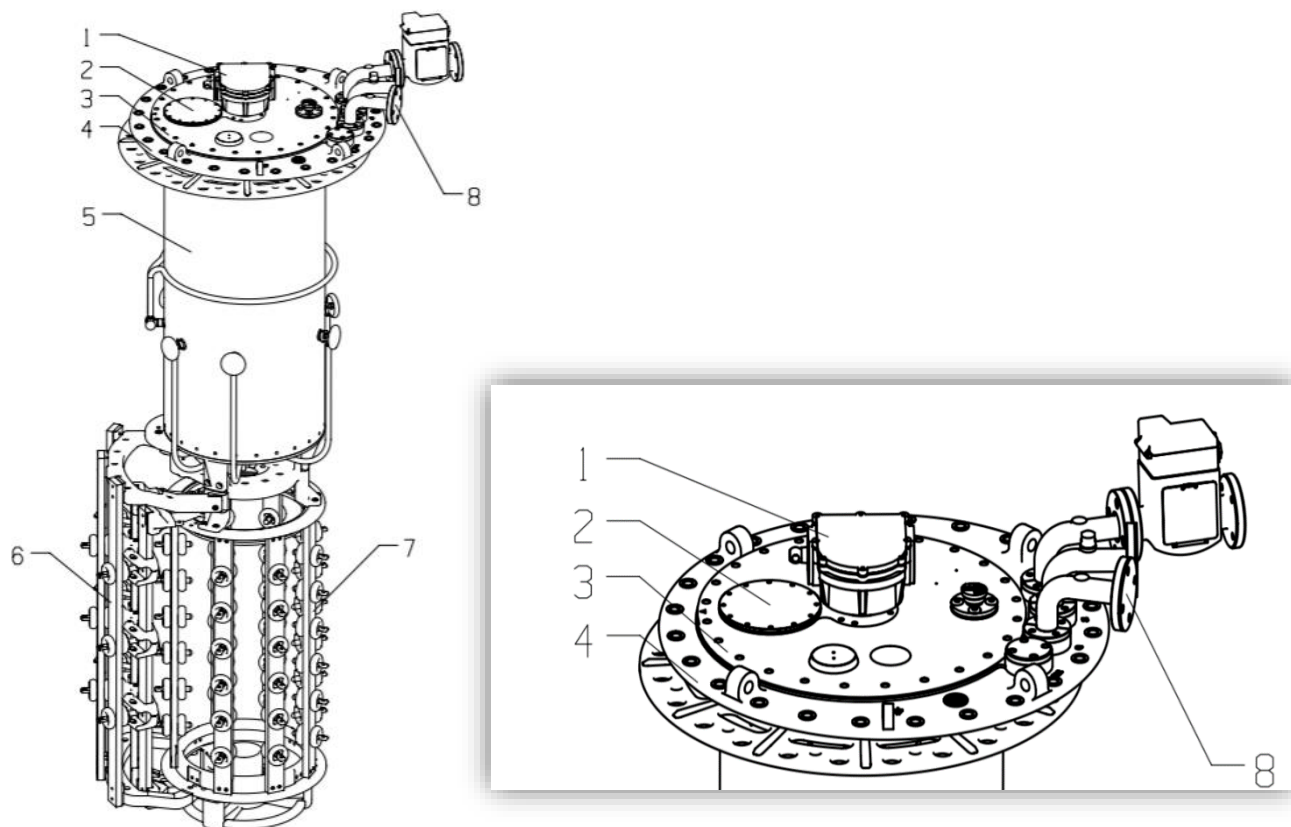


Figure 2 Overview of tap-changer

1. Upper gear unit	2. Rupture disk
3. Tap-changer head cover	4. Tap-changer head
5. Diverter switch	6. Change-over selector
7. Tap selector	8. Pipe bend

Table 4 List of tap-changer components

3. Technical data

CM tap-changer complies with IEC 60214-1: 2014 standard.

3.1 Table of technical data

Description		CM III 500Y CMII-500 CMI-500	CM III 600Y CMII-600 CMI-600	CMI-800	CMI-1200	CMI-1500
Max. rated through current (A)		500	600	800	1200	1500
Rated frequency (Hz)		50 or 60				
Connection		3-phase Y-connection for neutral point only Single-phase for any selectable winding connection				
Max. rated step voltage (V)		3300				
Rated step capacity (kVA)		1400	1500	2000	3100	3500
Short-circuit current test (kA)	Thermal (3s)	8	8	16	24	24
	Dynamic (peak)	20	20	40	60	60
Max. operating positions		See Figure 6				
Insulation to ground	The highest voltage for equipment (kV)	72.5	126	170	252	300
	Rated separate source AC withstand voltage (kV/50Hz, 1min)	140	230	325	460	480
	Rated lightning impulse withstand voltage (kV,1.2/50μs)	350	550	750	1050	1100
Tap selector		Categorized into B, C, D & DE four sizes				
Mechanical life		Not less than 800,000 operations				
Electrical life		Not less than 200,000 operations				
Oil compartment of diverter switch	Service pressure	0.03MPa				
	Leakage test	No leakage under 0.1 MPa for 24 hours				
	Over pressure protection	Rupture disc bursts at $300 \pm 20\%$ KPa				
	Protective relay	Set oil flow speed at $1.0\text{m/s} \pm 10\%$				
Motor drive unit		CMA7 or SHM-D				

Note : The step capacity is the product of the step voltage and the load current, and the rated step capacity is the maximum step capacity allowed continuously.

Table 5 CM technical data

3.2 Operation conditions

- 1). The operating temperature of the tap-changer in oil is $\geq -25^{\circ}\text{C}$ and $\leq 105^{\circ}\text{C}$.
- 2). The storage environment temperature of the tap-changer is $\geq -25^{\circ}\text{C}$ and $\leq 40^{\circ}\text{C}$, and the humidity should be not higher than 85%. If beyond this standard, please mark your requirements before ordering. The tap-changer material and accessories will be specially processed to meet the special requirements of the environment.
- 3). After being mounted to the transformer, the non-perpendicular degree between the tap-changer and the ground should be less than 2%.
- 4). No sever dust and other explosive or corrosive gases on the installation site.

**CAUTION!**

Do not store the tap-changer or motor drive unit in a place with high humidity ($\geq 85\%$). This will cause condensation.

**NOTE!**

Special design is available to meet particular environment conditions. Please contact Huaming for further information.

4. Type designation

4.1 Type designation

CM tap-changers have a variety of specifications due to different combinations of the number of phases, the maximum rated through current, the highest voltage of the equipment, the insulation class of the selector and the connection mode. Therefore, the above-mentioned performance parameters should be clearly reflected in the model of the tap-changer, and the detailed description of each code is shown in Figure 3.

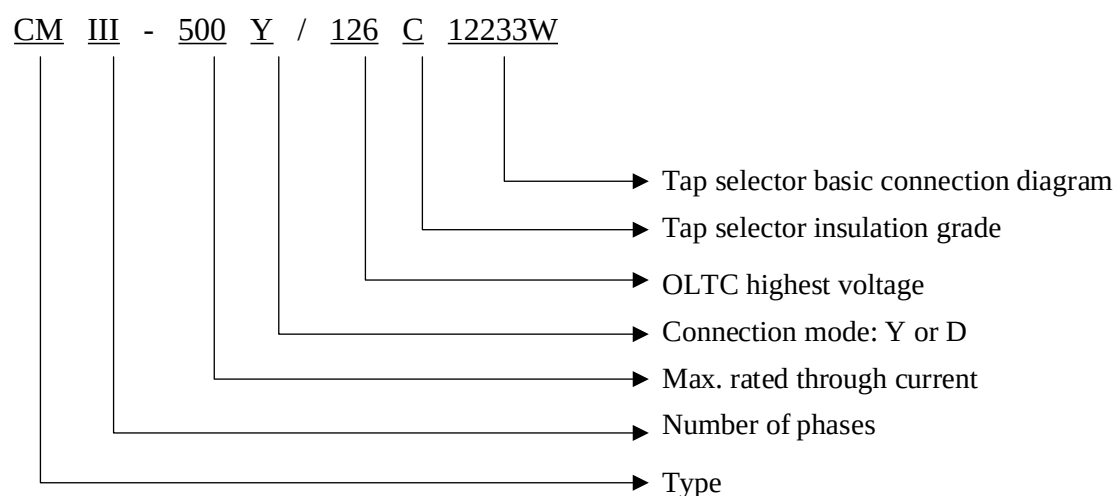


Figure 3 CM model designation

4.1.1 Basic connection

According to the transformer voltage regulation and winding connection mode, the tap selector has a variety of different specifications. It is composed of the number of tap selector pitch, the number of operating positions and the form of the changeover selector. The designation of the tap selector model is shown in Figure 4.

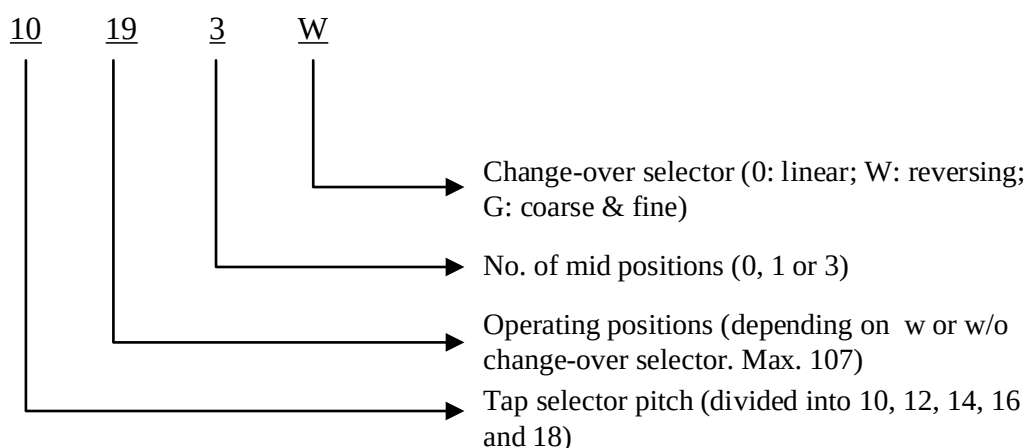


Figure 4 Tap selector basic connection

For example:

- 1). Linear regulation: It is represented by 5 digits. For example, 14140 means that the contact position of each phase is 14, the number of working positions is 14, and the number of intermediate positions is 0.
- 2). Reversing regulation: Add a letter W after 5 digits, such as 14131W, which means that the contact position of each phase is 14, the number of working positions is 13. The number of intermediate positions is reversing adjustment.
- 3). Coarse and fine regulation: 5 digits followed by a letter G, such as 14131G, indicating the contact position of each phase 14, the number of working positions is 13. The number of intermediate positions is coarse and fine adjustment.

4.2 Insulation level of tap selector

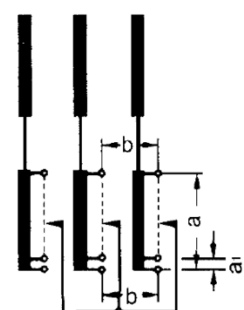
The tap selector has 4 different insulation classes, which are represented by B, C, D and DE respectively. The insulation class data is shown in Table 6, and the insulation distance symbol description is shown in Figure 5.

unit(kV)

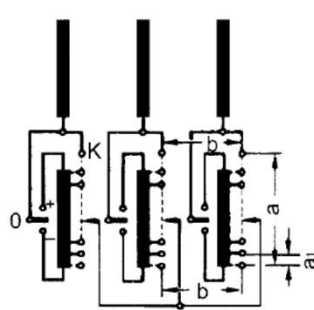
Insulation distance	Tap selector B		Tap selector C		Tap selector D		Tap selector DE	
	1.2/50 μ s	50Hz/1min	1.2/50 μ s	50Hz/1min	1.2/50 μ s	50Hz/1min	1.2/50 μ s	50Hz/1min
a	265	50	350	82	460	105	550	120
b	265	50	350	82	490	146	550	160
a0	90	20	90	20	90	20	90	20
a1	150	30	150	30	150	30	150	30
c1	485	143	545	178	590	208	660	230
c2	495	150	550	182	590	225	660	250
d	265	50	350	82	460	105	550	120

Note: a0, the internal insulation level is the insulation level with spark gap protection, when the full wave impact is 130kV, the spark gap will response 100%.

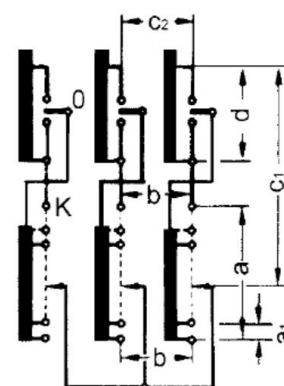
Table 6 Insulation level of Tap selector



10 10 0 ... 18 18 0



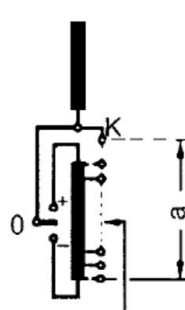
10 09 1W ... 18 17 1W
10 19 1W ... 18 35 1W
10 19 3W ... 18 35 3W



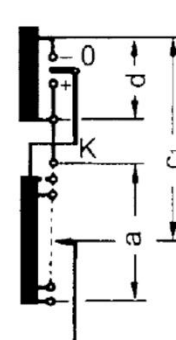
10 19 1G ... 18 35 1G
10 19 3G ... 18 35 3G



10 10 0 ... 18 18 0



10 09 1W ... 18 17 1W
10 19 1W ... 18 35 1W
10 19 3W ... 18 35 3W

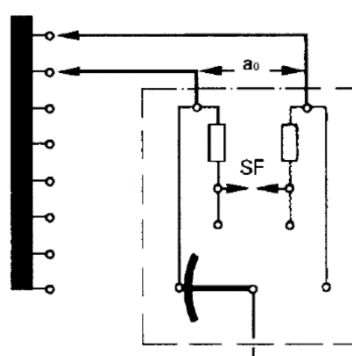


10 19 1G ... 18 35 1G
10 19 3G ... 18 35 3G

1. Linear

2. Reversing

3. Coarse & fine



4. Diverter switch

Explanation of designation code:

a: across regulation winding for the same phase

a1: between any selected and preselected taps of the tap selector

b: between any two taps of different phases

a0: between any adjacent taps of diverter switch

c1: between the beginning of coarse tap winding and neutral of fine tap winding of the same phase

c2: between beginnings of coarse tap winding of different phases

d: between beginning and end of coarse tap winding of the same phase

SF: spark gap

Figure 5 The voltage gradient on different regulating windings

4.3 Tap selector wiring diagram

The different transformer voltage winding tapping methods correspond to different specifications of tap selector's basic wiring diagrams. Figure 6 shows the common basic wiring diagrams. Special designs can be made to meet customers' requirements.

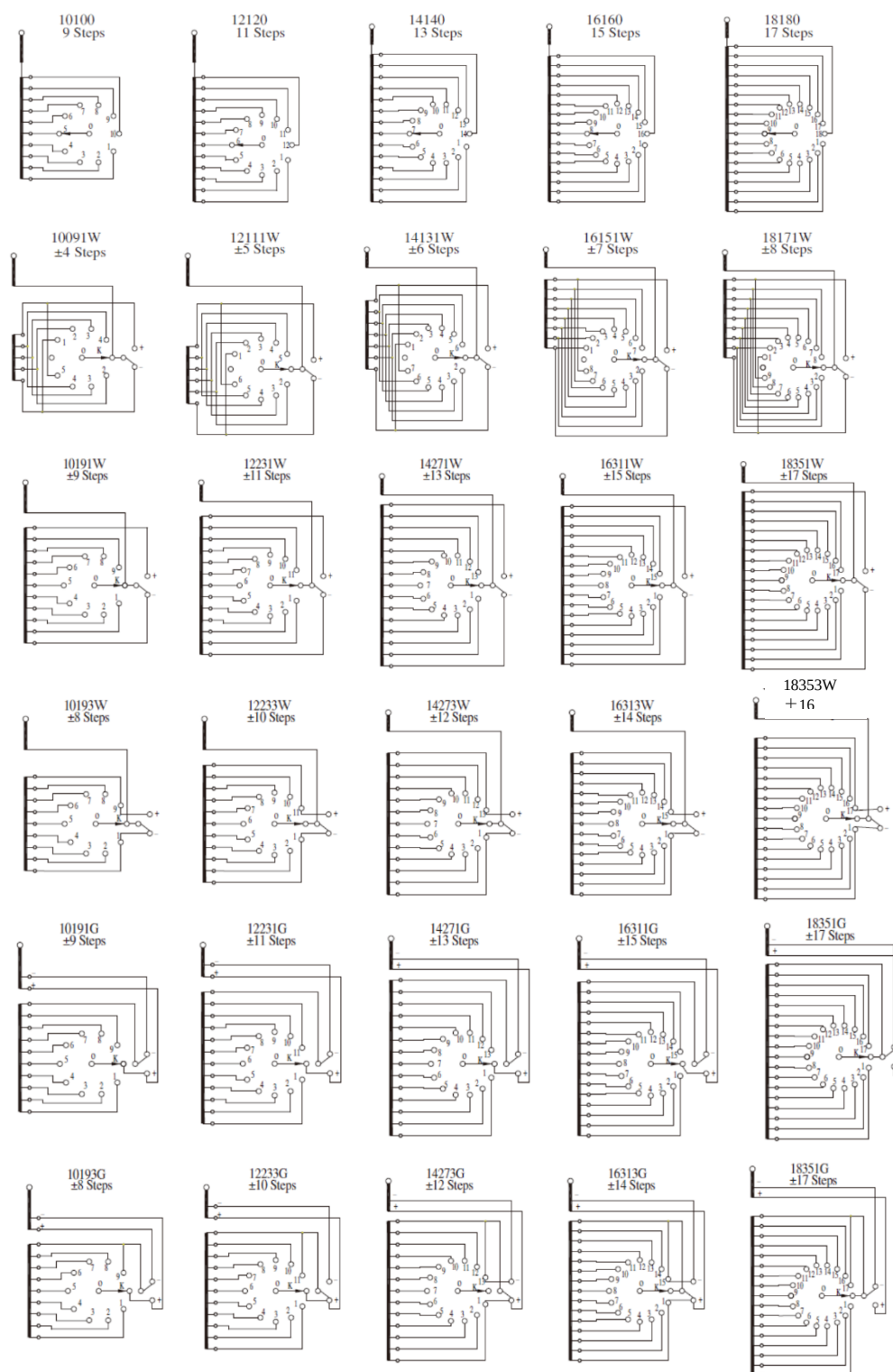


Figure 6 Basic connection diagram of Tap selector

5. Packing, transportation & storage

5.1 Supplied components

The complete set of tap-changer contains below parts (Figure 7):

- Diverter switch
- Tap selector
- Motor drive unit (MDU), controller and connecting cables, transmission shafts, couplings, bevel gear boxes and other transmission accessories
- Protection relay and accessories
- Special tools and accessories for tap-changer assembly
- Customer-required special accessories if any.

Please refer to the corresponding packing list delivered with the tap-changer for exact supplied components.



Figure 7 Set of tap-changer

5.2 Transportation & acceptance

5.2.1 Transportation

- Packings is suitable for all common kinds of transportations. The boxes of tap selector and MDU are allowed to be appropriately stacked during storage, and the load-bearing capacity of the box's top surface shall not exceed 500kg/m².
- Boxes should be placed reasonably according to the position of the gravity center. When lifting it up, the markings on the box must be observed (Figure 8).



Figure 8 Example of markings on the box

**DANGER!**

When lifting and storing the boxes, people are not allowed to stand under or near the boxes in case the boxes drop. All boxes must be stored

**CAUTION!**

Choose appropriate forklift and crane according to the dimension and gross weight of the boxes.

5.2.2 Goods receiving

Each batch of delivered goods must be checked by the consignee before accepted (signing acceptance confirmation). Following items must be checked by the consignee carefully:

- When receiving the goods, check the quantity of components received according to the packing list.
- Check whether the packing is intact. If it's found to be damaged (if possible, take pictures of the box and damaged goods) or the information of the goods does not match the packing list, consignee should refuse to sign for acceptance, and immediately report the problem to the shipping company and Huaming.
- The above steps are also applicable to corrosion of the goods due to moisture immersion (rain, snow, condensed water). The components must be stored in a dry place before installation.
- The on-load tap-changer must be stored in a sealed packing box and opened before installation only.

**NOTE!**

When receiving the goods, please check carefully the packing and the supplied components according to the packing list. In case anything is broken or missing, please contact Huaming in time.

**NOTE!**

The transportation and lifting of the box must be operated by professionals. The carrying capacity of transportation vehicles and cranes should be more than 1000kg.

**CAUTION!**

The packing box should be firmly fixed during transportation and lifting, and be handled with care to avoid possible damage to the product due to vibration, impact, falling, tipping, bumping, shaking, etc.

**CAUTION!**

If the box drops or is severely impacted, the possibility of damages must be considered. Comprehensive inspection of the product should be carried out, and if necessary, inspection(repair) shall be done by the Huaming's professional and technical personnel.

5.2.1 Acceptance

Unpack the box and look for damage during transportation.

- Transport the box to the installation site.
- Check the equipment and its accessories according to the packing list.

**CAUTION!**

During unpacking and inspection, avoid damaging the original package.

**NOTE!**

If the received goods and its accessories do not match the list, contact Huaming in time.

5.3 Storage

When the goods are stored continuously for more than 1 year, the equipment should be inspected thoroughly before installation.

Specially packed goods with sealed packing can be stored outdoors if they meet the following conditions.

- The storage environment temperature should not be $\geq -25^{\circ}\text{C}$ and $\leq 40^{\circ}\text{C}$.
- The tap-changer must not be stored in an environment with flammable, explosive and corrosive gases.
- It must be protected from moisture, dust, rodents and ants.
- Regularly check whether the stored equipment is abnormal
- If the tap-changer is stored for a long time, the desiccant must be replaced regularly, and re-packed with sealed packing.

**CAUTION!**

After long-term storage, the tap-changer and its accessories must be carefully inspected before installation.

6. Mounting of tap-changer on the transformer

6.1 Transformer mounting flange for tap-changer installation

A mounting flange is required to fix the tap-changer head on the transformer cover. The structure of the flange should be designed according to the shape of the tap-changer head's sealing surface (Figure 9). Double-thread screw (M12, maximum length=45mm) should be accurately positioned.

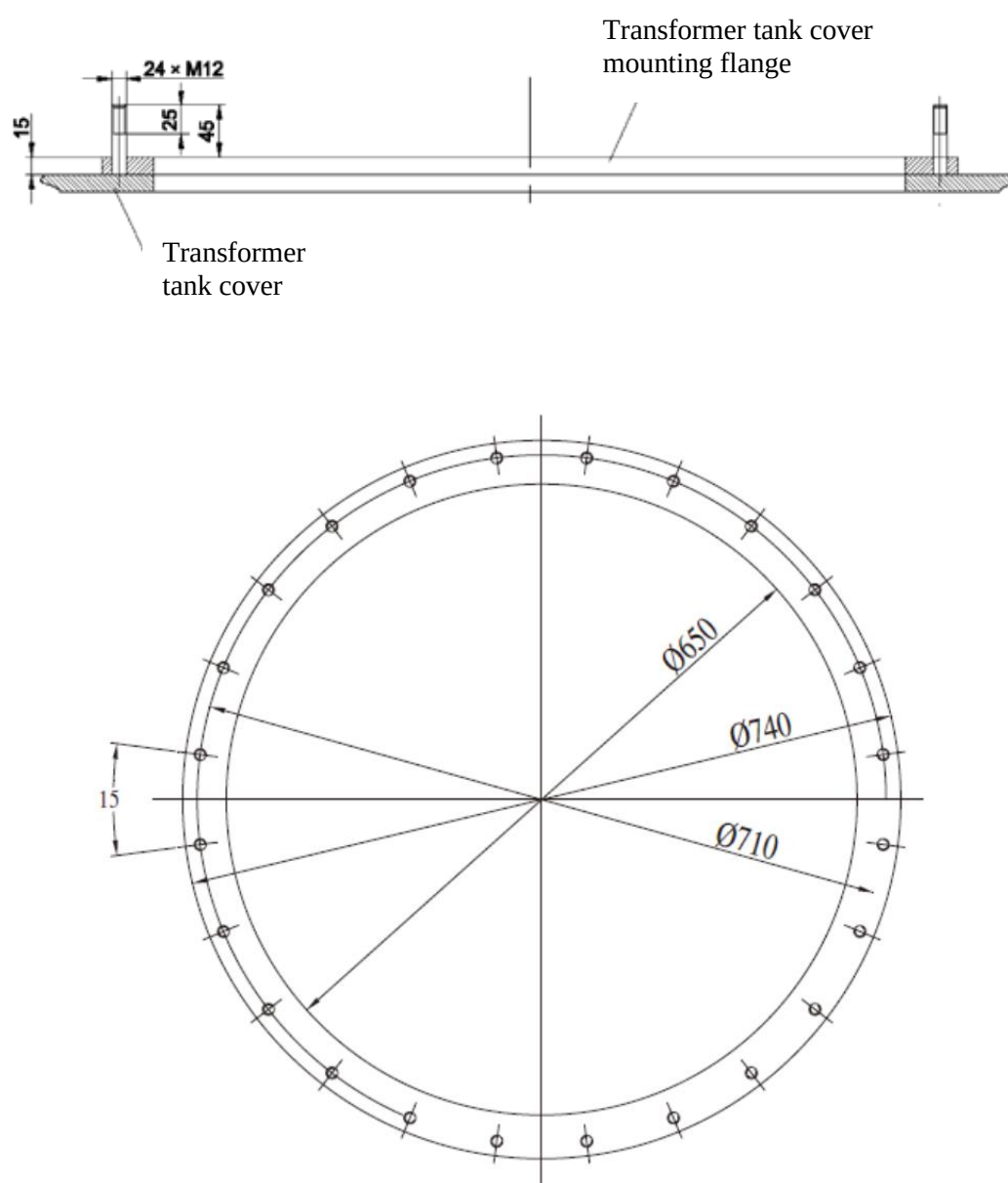


Figure 9 Transformer mounting flange

6.2 Assembly of tap selector and diverter switch



DANGER!

If heavy components are not placed on a flat surface, they may tilt and cause injuries! Therefore, they must be put on a horizontally flat surface.



CAUTION!

If heavy components are not placed on a flat surface, they may tilt and cause damages to the products! Therefore, they must be put on a horizontally flat surface.

Lift the diverter switch oil compartment to be above the tap selector and connect them as follows. For bell-type tap-changer, first remove its head cover and then lift out the switch insert.

- 1) Place the tap selector on a horizontal flat table. Remove tap selector's 6 pieces of M12 hexagon socket bolts (8mm wrench, Figure 10), spring washers, nuts, etc., and store them properly for use later. Then remove the positioning pin on the tap selector (Figure 11).

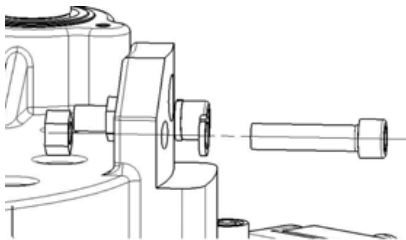


Figure 10 Remove hexagon socket bolt

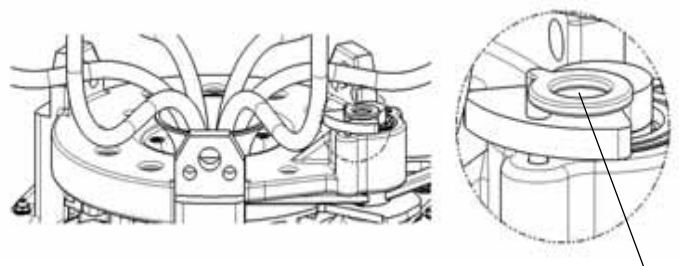


Figure 11 Remove the positioning pin

Remove

- 2) Lift the oil compartment and remove the positioning pin on diverter switch (Figure 12).

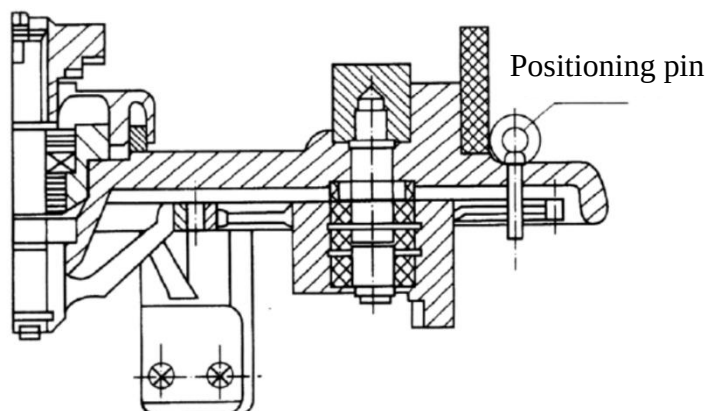


Figure 12 Remove the positioning pin on diverter switch

- 3) Place the oil compartment carefully to be above the selector. Keep sufficient distance between lead wire of the tap selector and the bottom of the diverter switch to protect the lead wire from damage.
- 4) Align the positions of two couplings and prevent the convex upper and lower connecting pieces from bumping into each other.

**CAUTION!**

Do not turn the coupling after removing the positioning pin.

- 5) Fix the tap selector to the bottom of the oil compartment with bolts: 6 pieces of M12 hexagon socket bolts (8mm wrench). Torque of bolt head, outer gasket and inner nut is 50 to 60N.m. (Figure 13)

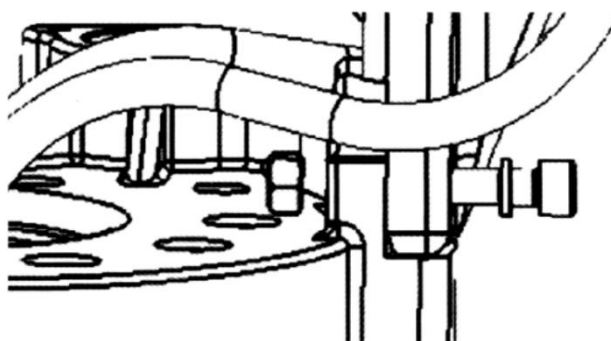


Figure 13 Tighten the bolt

**CAUTION!**

Pay attention to the perpendicularity between the diverter switch and tap selector (error less than 2°).

6.3 Connect the lead wire of the tap selector to the diverter switch oil compartment

- Remove the wiring bolts from the contacts of the oil compartment. The order of fasteners from outside to inside: wiring bolt, spring washer, shield cap, tap selector lead wire.
- Fix the lead wire of tap selector on the connecting contact of oil compartment.
- Two connecting wires must be connected to each phase of the diverter switch (six wires have been fixed to the selector before delivery). Tighten the 6 pieces of M10 hexagon bolt (8mm wrench), torque 28 to 34 N.m. (Figures 14&15)

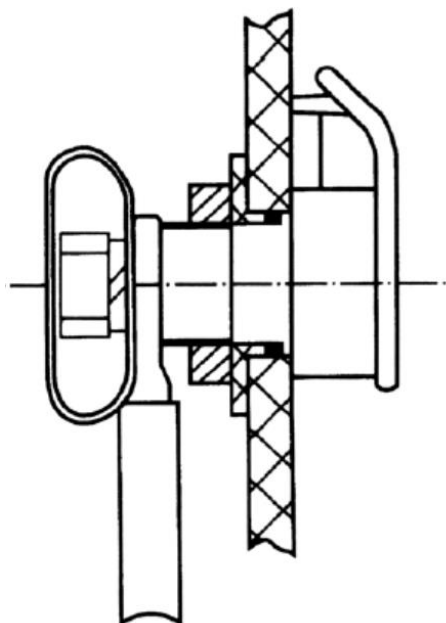


Figure 14 Contact

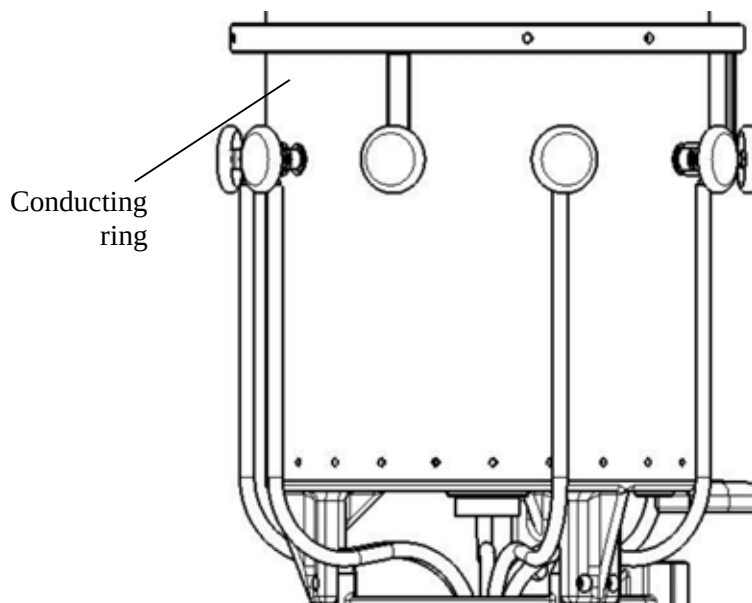


Figure 15 Conducting ring



CAUTION!

Align the wire directly with the contact on the oil compartment. Do not make the shielding cap between them (Figure 16).



CAUTION!

All lead wires must be connected carefully. The specified tightening torque must be used, the bolt connection must be completely reliable, and the terminal part must be shielded with the shielding cap provided together with the tap-changer.



DANGER!

When connecting tap selector and diverter switch, be careful not to damage the outer insulation layer of the wire. When working on motor drive unit or components of tap-changer, make sure that they are all firmly fixed! Prevent parts of device from accidentally toppling over!

- 1) III-500/600 & I-1200/1500 tap selector's lead wires (6 in total, Figure 15) must be connected to the diverter switch contact led-out terminal (with holes for M10, Figure 14).
- 2) II-500/600 & I-800 tap selector's lead wires (4 in total) must be connected to the diverter switch phases B and C's contact led-out terminal. Then use a wire to short-circuit phase A (Figure 16).

**NOTE!**

Only single-phase tap-changers use conducting rings, including I-500, I-800, I-1200/1500.

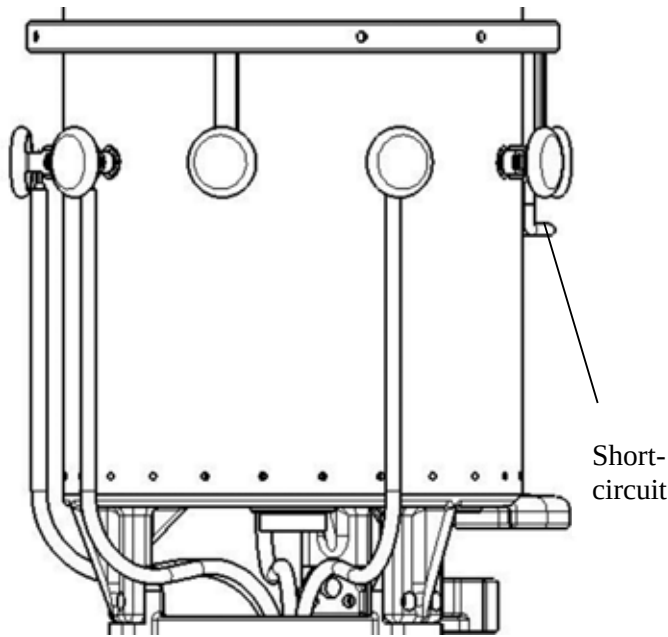


Figure 16 Short-circuit phase A

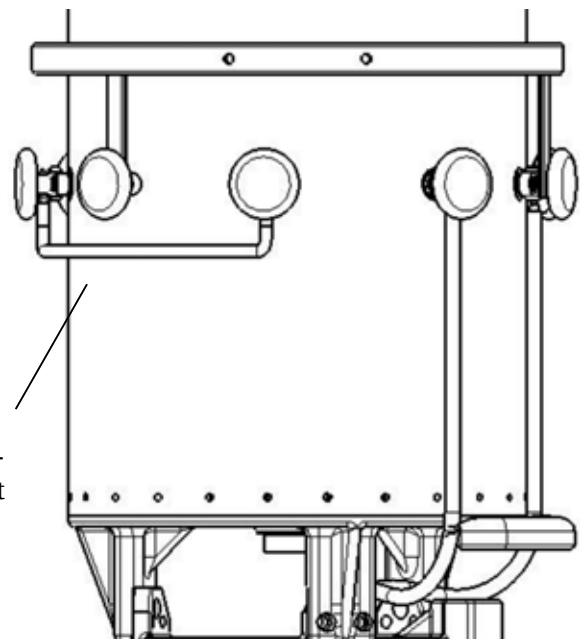


Figure 17 Short-circuit phases B and C

- 3) I-500/600 tap selector's lead wires (2 in total) must be connected to the diverter switch phase A's contact led-out terminal. Then use a wire to short-circuit phases B and C (Figure 17).
- 4) The shielding caps must be fastened after bolt connection.

**CAUTION!**

When it comes to multiple coarse & fine regulation tap selector, lead wires must be carefully connected to coarse & fine selector's wiring terminals. The distance between these wires and the adjacent wiring terminals should be as large as possible. In order to ensure the withstand voltage strength, it is recommended wrapping 3mm insulation paper on the lead wires where terminals of the coarse & fine selectors are too close. (Figure 18)

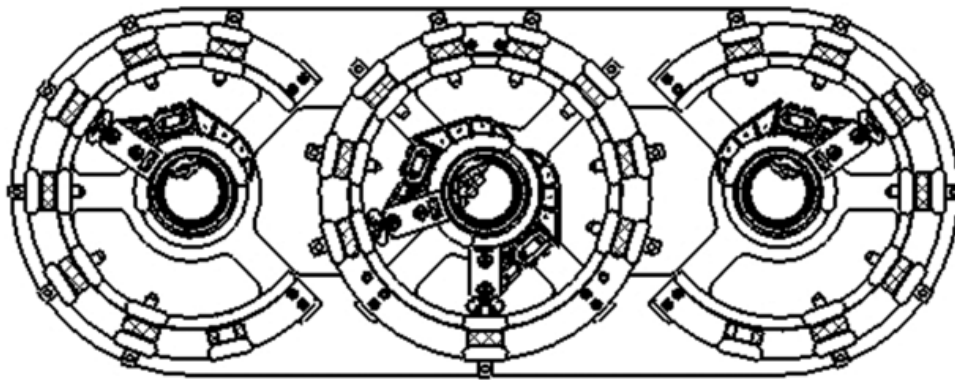


Figure 18 Wrapping insulation paper

**NOTE!**

1. Be careful with the connection of tap selector.
2. Use specified tightening torque.
3. Make sure the bolt connection is reliable.
4. Screen caps to shield the terminal interface.

6.4 Mounting of tap-changer on standard type transformer

The mounting steps are as follows:

- 1) Clean the bottom sealing face of the tap-changer head flange as well as the surface of transformer mounting flange, and place a sealing gasket on it (Figure 19).
- 2) Lift the diverter switch to be above the transformer's mounting flange. Carefully place it into the transformer tank through the mounting hole on the transformer cover (Figure 20).

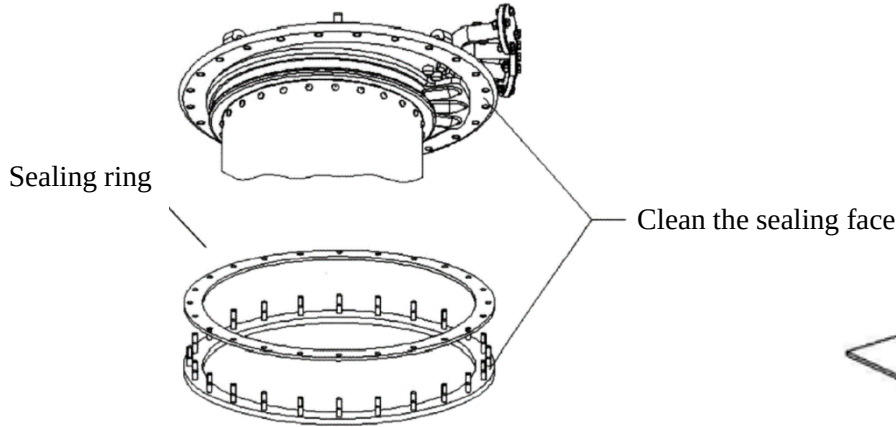


Figure 19 Sealing face

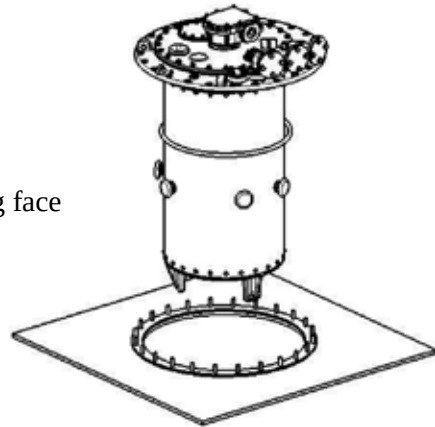


Figure 20 Mounting hole

- 3) Check whether the tap-changer head's mounting position and angle meet the design requirements. If yes, fix the tap-changer head flange to the transformer's mounting flange, and fasten 24 pieces of nuts with a torque of 100 to 110 N.m.
- 4) Assembly the tap selector with the diverter switch.



CAUTION!

Make sure vertically lower the oil compartment into the assembly hole, so as not to damage the tap-changer's wiring terminal and the diverter switch's shield cap and grading ring (grading ring is applied only when the voltage is 170kV and above).

6.5 Mounting of tap-changer on bell-type transformer



NOTE !

1. Make sure small parts do not fall into OLTC oil compartment.
2. During disassemble and reassemble, all small parts must be complete. Make sure to check carefully.

6.5.1 Removing OLTC assembly flange and lifting out the switch insert

- 1) Make sure the tap-changer is at the calibration position.
- 2) Unscrew the 24 pieces of M10 bolts on the tap-changer head cover (with spring washer, 17mm wrench).
- 3) Remove the tap-changer head cover.

During disassembly and other operations, avoid damaging the head cover and the sealing surface (O-ring) (Figure 21).

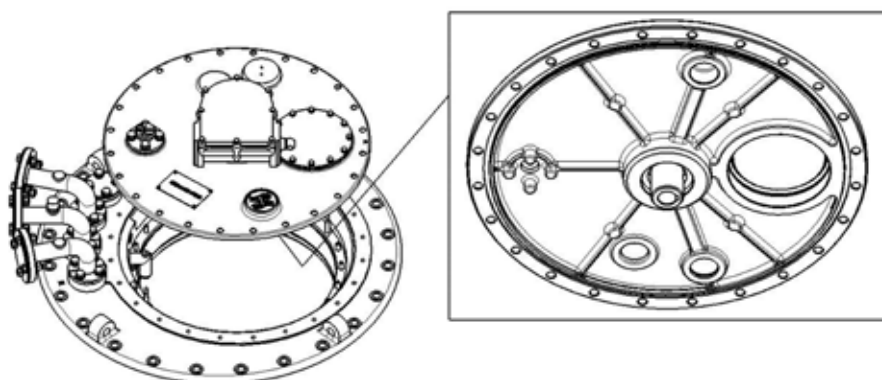


Figure 21 Tap-changer head cover

- 4) To remove the position indicating disc, pull out the open locking tab from the shaft end and disassemble the disc (Figure 22).

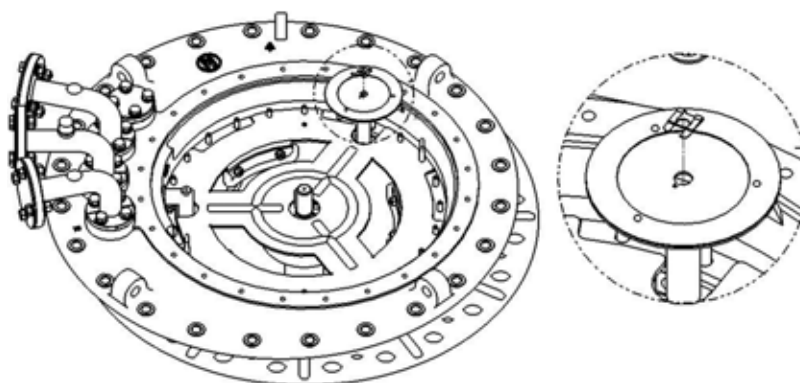


Figure 22 Remove the positioning indication disc

- 5) Unscrew the fastening nuts on the support plate (not red area) (10 pieces of M8 nuts, 13mm wrench, with spring washer).
- 6) Carefully lift out the switch insert vertically from the oil compartment (Figure 23).

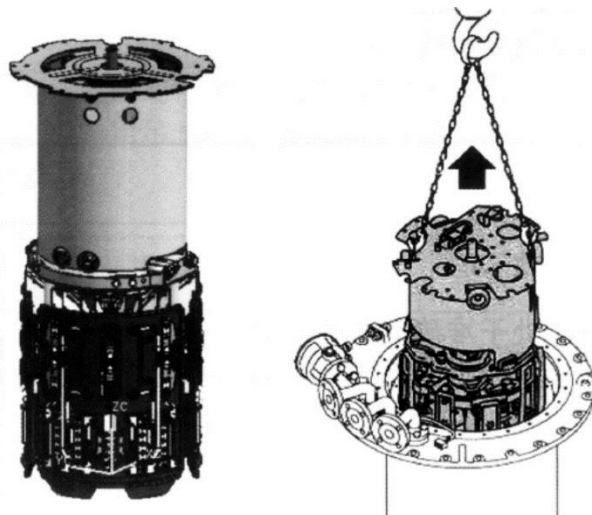


Figure 23 Lift out

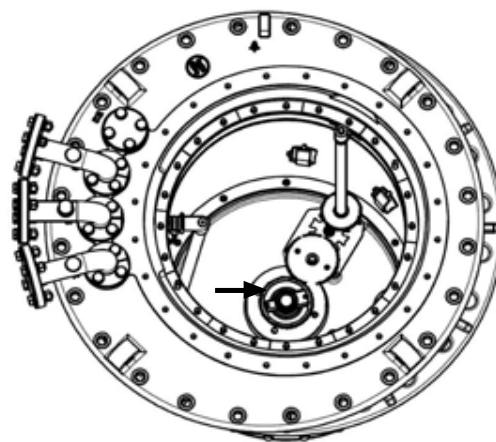


Figure 24 Remove the oil suction pipe switch insert

- 7) Pull out the pipe joint inserted in the tap-changer head flange, and disassemble the oil suction pipe (Figure 24).

**CAUTION!**

Don't damage the O ring on the pipe joint

- 8) Remove the remaining fixing nuts on the mounting flange (12 pieces of M8 nuts, 13mm wrench with spring washer). Lift the tap-changer mounting flange from the supporting flange.

**CAUTION!**

Do not damage the O ring.
Do not drop fasteners into the oil compartment. Keep them in safe place.

**CAUTION!**

After lifting out the switch insert, do not change the tap selector position during assembly.

**DANGER!**

Put the switch insert in a flat place in case it tilts, which may cause damage to the device itself and to personnel on site.

6.5.2 Installation of the tap-changer into the bell-type transformer tank

**CAUTION!**

1. The tap-changer must be in the accurate vertical position in terms of the support structure. (Max. 2° vertical position deviation). Otherwise, tap windings connecting the OLTC led-out terminals may apply force on the OLTC, causing damages. What's more, tap selector contacts may be wrongly connected.
2. The installation of the tap-changer supporting structure must ensure that the tap-changer can be lifted 5 to 20 mm to its final position after the bell-type transformer tank cover is closed.

Place the tap-changer on the temporary support frame inside the transformer. For this purpose, the tap-changer is equipped with a support flange on the oil compartment. The assembly steps are as follows:

- 1) Lift the tap-changer onto the support frame and connect it to the voltage regulating winding. The tap-changer must be stably placed on the frame and is not allowed to move randomly.
- 2) Use the mounting hole on the support flange to temporarily fix the tap-changer on the frame. If necessary, put a temporary adjusting block between the temporary support frame and the support flange. Adjust the installation height of the tap-changer to the final installation position. The adjusting block must be removed before installation of the bell-type transformer tank cover. (Figure 25)

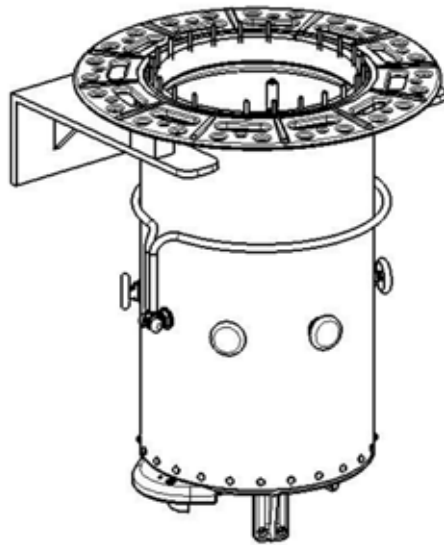


Figure 25 Putting tap-changer onto the support frame

**CAUTION!**

The connected tap terminal wiring must not cause any force to the tap-changer. Enough space should be left so that the tap-changer can be lifted to the final position after the bell-type tank is in place.

6.5.3 Assembly of bell-type transformer tank cover

- 1) Before assembling the bell-type transformer tank cover, clean the sealing surface of the tap-changer's support flange. Place the sealing ring on the support flange (Figure 26), and pull out the adjusting blocking.

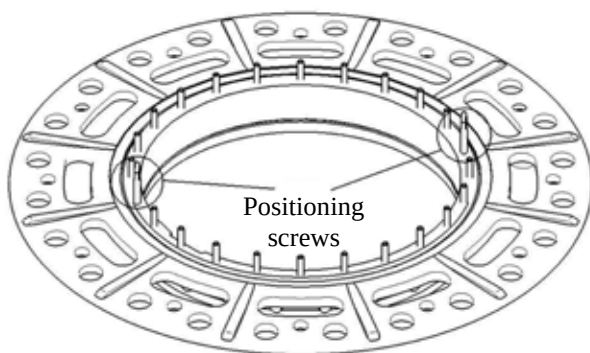


Figure 26 Sealing ring on the support flange

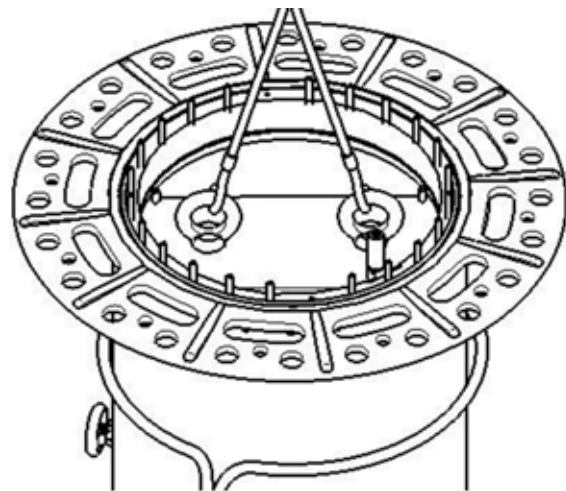


Figure 27 Lift the tap-changer

- 2) Lift the bell-type transformer tank cover over the transformer body and assemble it.
- 3) Clean the sealing surface before installing the tap-changer mounting flange. Place the sealing gasket on the transformer mounting flange, and fix the tap-changer mounting flange on the transformer's (24 pieces of M10 bolts, torque 100 to 110 N.m.).
- 4) Lift the tap-changer slightly using special slings. Make sure that all screws of the support flange are accurately positioned in the tap-changer fixing hole (Figure 27).



CAUTION!

Always use the professional special lifting tool to vertically lift up the OLTC. It's forbidden to use the supporting flange's connecting bolts for the lifting process. Otherwise, the bolts will be damaged.

- 5) Install the tap-changer on the head mounting flange. Pay attention to the two positioning screws (lengthened) and the red triangle marks on the support flange as well as on the tap-changer head cover, so as to ensure it is installed in the correct position.

6.5.4 Installation of diverter switch insert

- 1) When installing the switch insert, make sure its gear at the bottom and the tap selector are in the setting position (Figure 28).
- 2) Diverter switch should be in the setting mounting position (Figure 29).

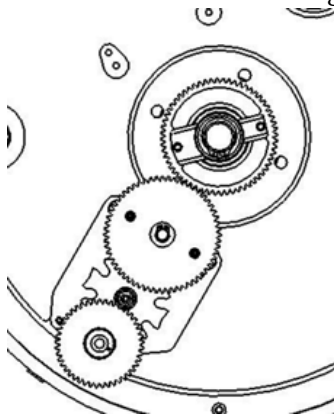


Figure 28 Setting position

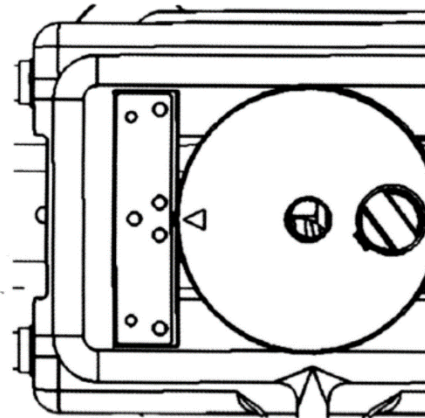


Figure 29 Diverter switch at the setting position

- 3) Install the oil suction pipe into the positioning hole at the bottom of the oil compartment (Figure 30). Then, insert the pipe into the tap-changer head flange mounting hole (Figure 31).

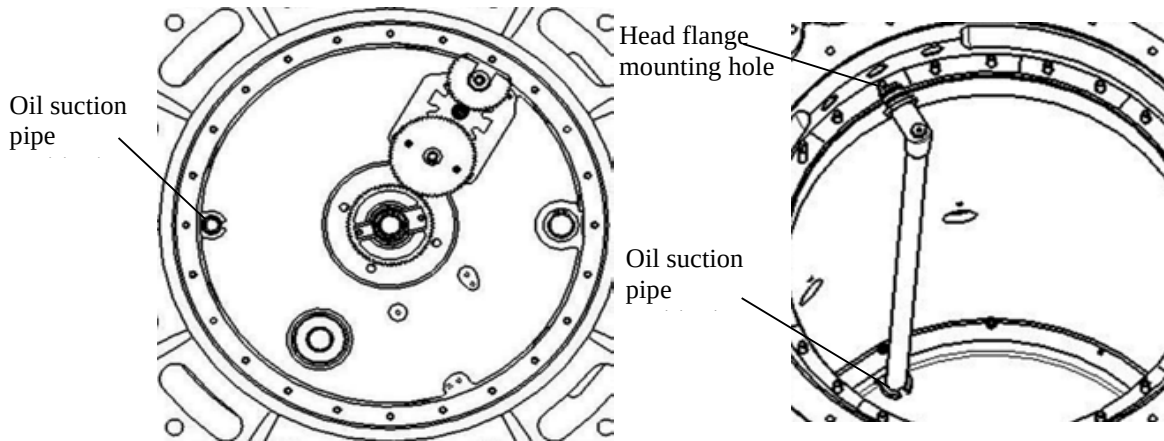


Figure 30 Install the pipe into the positioning hole Figure 31 Insert the pipe into the mounting hole



NOTE!

Make sure that the O-ring is placed in the correct position.
The O-ring seal should be slightly coated with lubricant before installation.

- 4) Lift the switch insert to be above the oil compartment, and slowly drop the insert to its final position. Make sure that the insert mounting plate in the compartment is in the correct position (switch insert's top plate has two gaps which is 120° away. These two gaps, one is aligned with the oil suction pipe and the other is aligned with the drive shaft used to install the position indicating disc). Fasten the support plate: 10 pieces of M8 nuts (13mm wrench with spring washer, torque 12 to 15 N.m.) (Figure 32).

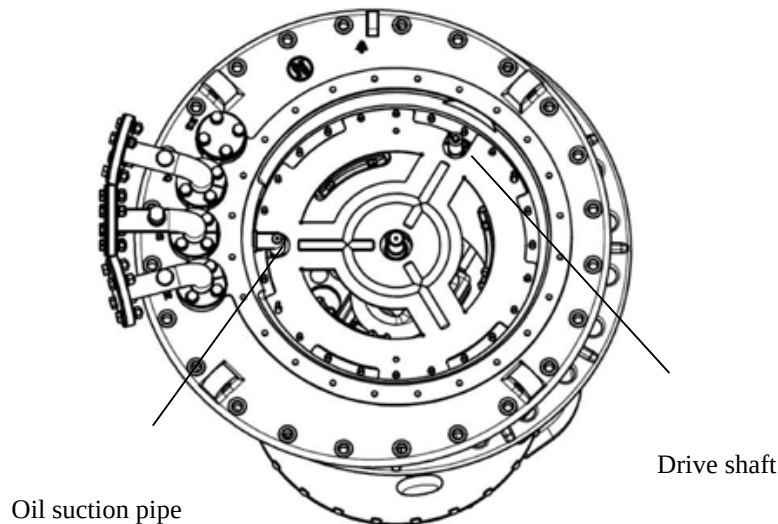


Figure 32 Install switch insert

- 5) Use nuts to tighten screws in the red marked area not covered by the diverter switch's support flange, and fasten the tap-changer head to the mounting flange with 12 pieces of M8 nuts (13mm wrench with spring washer, torque 12 to 15 N.m.) (Figure 33)



Figure 33 Tighten screws in the red area

- 6) Reinstall the position indicating disc and make sure the snap rings are fastened in the right place.
7) Place the tap-changer head cover on the head flange to close it, making sure that the sealing gasket in the cover is in the correct position.

**CAUTION!**

The red marks on the head cover and the flange should be aligned. Wipe the sealing surface carefully before installing the head cover (Figure 34).



Figure 34 Align the two red marks

- 8) Fasten the tap-changer head cover: evenly fasten 24 pieces of M10 bolts (17mm wrench), torque is 29 to 35 N.m.

6.6 Connection between tap of voltage regulating winding and terminal of tap-changer

The tap of the voltage regulating winding must be connected according to the wiring schematic diagram. The terminal of the tap-changer shall be marked with the contact position mark on the insulating rod of the tap selector. The tap of the voltage regulating winding and the terminal of the tap-changer shall be connected correctly according to the marking code on the tap-changer.



CAUTION!

The wiring of all tapping leads connected to the tap-changer must be reliable and tight. The assembly of tap lead shall be such that the lead connected to the tap selector does not pull the terminal.

6.6.1 The tap of voltage regulating winding fixed with tap-changer's terminal

The tap selector terminal is provided with a through hole for the M10 bolt, which is convenient for the connection and fixation between the tap of voltage regulating winding and the terminal of tap-changer.

After tightening it by the M10 connecting nuts which is provided with tap-changer, the washer of shield cover should be turned up 90° in two sides. (Figure 35)

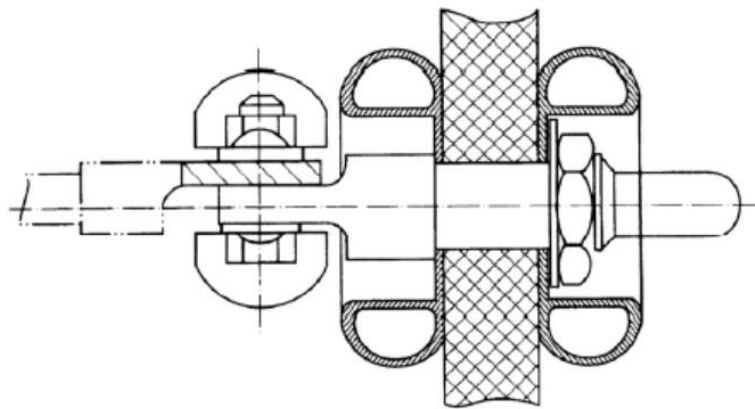


Figure 35 Washer of shield cover

6.6.1.1 Terminals of reversing tap selector

Reversing tap selector's (+) and (-) terminals as well as change-over selector's terminal K all have M10 bolt through-holes for connection and fixation of regulating winding's tap terminals (Figure 36).

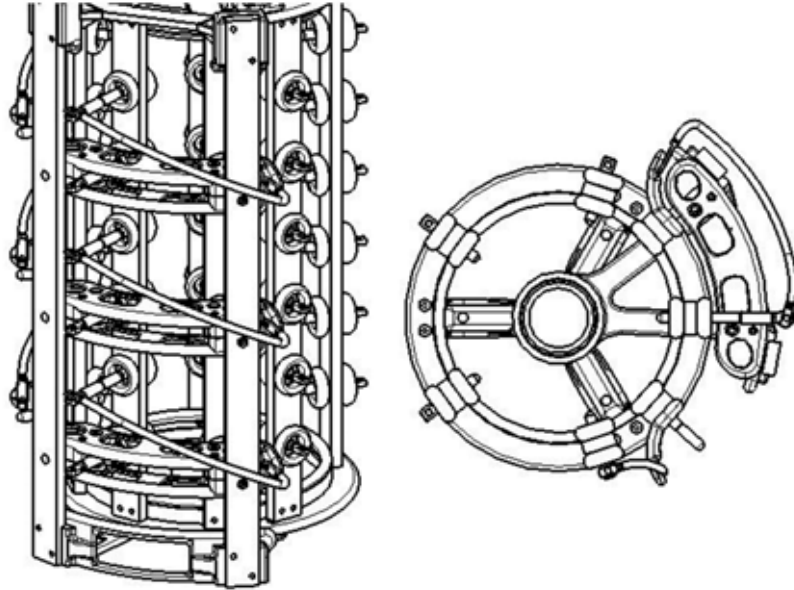


Figure 36 Tap selector



CAUTION!

The wire connected to terminal K must not be bent or deformed. Otherwise, the performance of the tap-changer will be affected.

6.6.1.2 Terminals of coarse & fine tap selector

Coarse & fine tap selector's terminals (+), (-) and (0) are similar to those of reversing tap selector. Therefore, the connection method is the same (Figure 37).

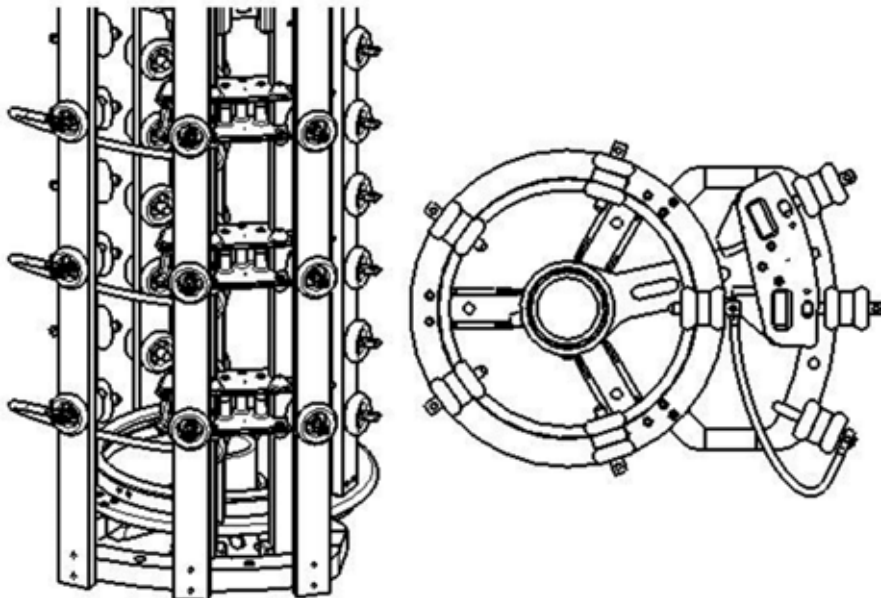


Figure 37 Coarse & fine tap selector

**CAUTION!**

The conductors close to the change-over selector must leave sufficient space for the change-over selector's movable parts. If the distance between conductors and change-over selector movable parts is too small, it will cause change-over selector to get stuck and consequently damage the OLTC!

6.6.2 The terminal lead of tap selector shall not deform or damage the tap-changer

- 1) The lead wire of voltage regulating winding of transformer shall be led to tap selector from both sides to avoid distortion of tap selector due to stress on one side lead wire.
- 2) The connection between the tap selector terminal and the terminal clamp of the transformer lead should have a certain deflection, which should not be too short, and the connection should be soft. It is recommended that this section of connection should not be coated with insulating paint, so as to avoid hardening and deformation of the insulating rod after drying.
- 3) The lead end connecting the tap selector shall be connected according to the expanded ring shape (looping), so that the insulating board of the tap selector is not subject to tension.
- 4) The tap selector terminal lead shall be led out from the outside of the selector and shall never pass through the inside of the selector.
- 5) The terminal lead of the change-over selector should be led out from the outside of the change-over selector insulating rod, and there must be enough clearance between the lead and the insulating rod of the tap selector's moving contact to prevent the movement of the change-over selector from being blocked.
- 6) The bell-type tap-changer must be lifted 5 to 20mm after the lead is connected. For this reason, special attention should be paid to the tightness of the connecting lead. It is recommended that the intermediate flange be installed on the supporting structure, and the intermediate flange should be temporarily raised by using a pad until the final installation finishes. Then connect the lead (same as the actual installation situation). After the lead is completed, remove the pad and check the tightness of the lead and whether the tap-changer is under force.

7. Transformer ratio test and transformer DC resistance measurement

**CAUTION!**

Operation without oil will lead to damage of OLTC.

- Without oil, tap-changer cannot be operated more than 10 cycles before drying.
- For the first operation of tap -hanger after transformer being dried, make sure tap selector is fully immersed in oil and the diverter switch oil compartment is full of oil.

**CAUTION!**

Incomplete tap change operation can cause damage to the OLTC!

**NOTE!**

It is suggested that the ratio test and DC resistance measurement should be carried out before drying.

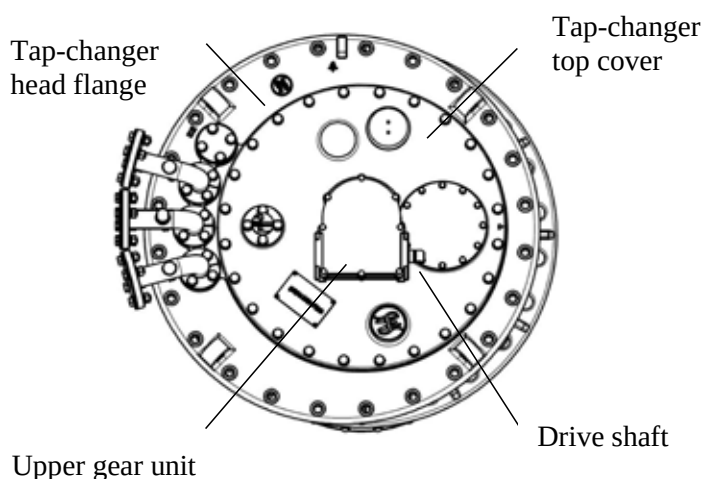


Figure 38 Tap-changer head

7.1 Transformer ratio test



NOTE!

For bell-type tap-changer, head flange, diverter switch and top cover need to be assembled temporarily.

When turning the drive shaft of the tap-changer upper gear unit (Figure 40), a short pipe with an inner diameter of 25 mm (a connecting bolt with a diameter of 12 mm has been screwed into the pipe) can be used, and then a hand wheel or hand crank is also needed.

For two or three sets of linkage switch group, all tap-changer heads should be connected with horizontal axis. For a tap change, the sound of the diverter switch action can be heard clearly. After the sound of the diverter switch be heard, it must continue to rotate two and a half turns in the same direction to complete one tap change. During operation, each operation position should be observed at any time through the observation window on the tap-changer head (Figure 39), so as to avoid exceeding the end position of the tap-changer.

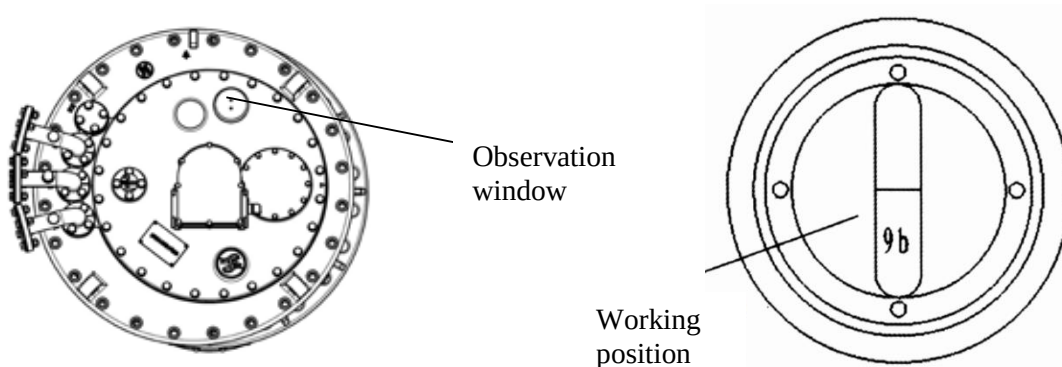


Figure 39 Observation window



CAUTION!

1. The tap-changer can only be operated through the transmission shaft of the gear box.
2. In the case of no oil, too much operation will cause damage to the tap-changer!
3. In the case of no oil, the tap-changer switching operation shall not exceed 10 cycles before drying.

7.2 Transformer DC resistance measurement

It should be noted that the tap-changer is operated without oil when measuring the DC resistance of transformer, and the operation of tap-changer should be reduced as much as possible.

After the transformer transformation ratio test and DC resistance measurement, the tap-changer must be adjusted back to the calibration position.

8. Drying process and oil filling

8.1 Drying process

To ensure the insulation property of tap-changer, necessary drying must be taken as per below description (vacuum drying or vapor phase drying).

8.1.1 Vacuum drying



CAUTION!

OLTC top cover, transmission gear box and OLTC accessories are not allowed to be vacuum dried together with the transformer in the drying oven. Otherwise, they will be damaged.

Before drying, OLTC top cover, transmission gearbox, MDU, protective relay, pressure relief valve, bevel gearbox, online oil filter and other accessories must be removed and cannot be dried.

8.1.1.1 Drying in the oven room

- 1) The tap-changer is heated up at a rate of 10°C/ hour in the air under atmospheric pressure, till the temperature reaches highest 110°C.
- 2) Drying is carried out in the circulating hot air for 20 hours and highest temperature of tap-changer is 110°C.
- 3) Vacuum drying lasts 50 hours, temperature of tap-changer is 105°C till 125°C. Maximum residual pressure 133Pa.

8.1.1.2 Drying in the transformer oil tank

- 1) By-pass pipe (nominal diameter 25mm) can be connected to the flange connection position between flanges E2 and Q.
- 2) The tap-changer is heated up at a rate of 10°C/ hour in the air under atmospheric pressure, till the temperature reaches highest 110°C.
- 3) Drying is carried out in the circulating hot air for 20 hours under max. tap-changer temperature 110°C.
- 4) Vacuum drying lasts 50 hours, temperature of tap-changer is 105°C till 110°C. Maximum residual pressure 133Pa.

8.1.2 Vapor phase drying



CAUTION!

OLTC top cover, transmission gear box and OLTC accessories are not allowed to be vacuum dried together with the transformer in the drying oven. Otherwise, they will be damaged.

Before drying process, tap-changer top cover, transmission gear box, motor drive unit, protective relay, pressure monitoring device, pressure relief device, bevel gearbox, temperature sensor, online oil filter and so on must be removed and placed outside the drying oven. In terms of vapor phase drying, be sure to open the oil drain stopper at the bottom of the oil compartment before drying, so that the condensate can be discharged from the oil compartment.

8.1.2.1 Loosening oil drain stopper

Use a special wrench to loosen the oil drain stopper counterclockwise. Complete loosening is not allowed. (Figure 40)

- 1) Turn clockwise to loosen the oil drain stopper between the oil compartment bottom and the tap selector. (Figure 41)
- 2) Remove the tap-changer top cover and store it outside the vacuum oven.
- 3) Remove all accessories except the tap-changer body and put them outside the drying oven.

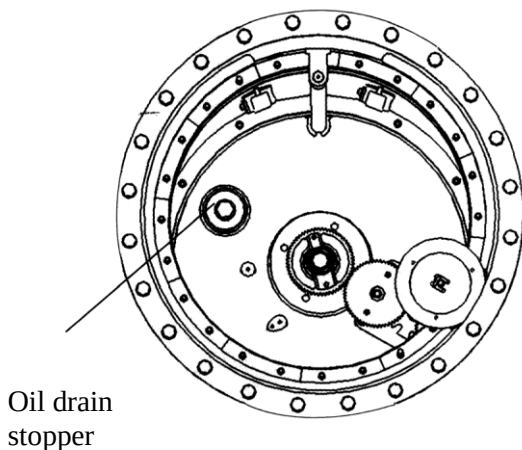


Figure 40 Loosen the oil drain stopper

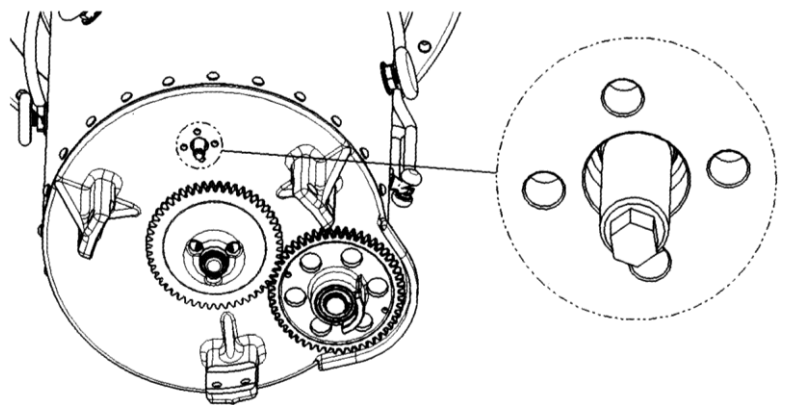


Figure 41 Loose the oil drain stopper
Between oil compartment & tap selector

- 4) Pass in kerosene vapor around 90°C and keep this temperature constant for 3 to 4 hours.
- 5) Increase the kerosene gas temperature at a rate of 10°C/hour till the specified final temperature, but maximum temperature of the tap-changer cannot exceed 125°C.
- 6) In vacuum drying, the temperature of the tap-changer is 105°C to a maximum of 125°C. The highest residual pressure is 133Pa. The drying duration is the same as the transformer, but at least 50 hours.
- 7) Close the oil drain bolt (torque 18~20N.m.) after drying process.

**CAUTION!**

Oil leakage will lead to OLTC and transformer damages! Loose oil drain stopper will cause oil to leak from the oil compartment which will damage OLTC and transformer.

8.1.2.2 Vapor phase drying inside the transformer tank

Before drying, the oil drain stopper at the bottom of the oil compartment must be loosened to discharge the condensate.

**CAUTION!**

Oil leakage will lead to OLTC and transformer damages!

**CAUTION!**

Keep all small components in a safe place and do not drop them inside the oil compartment.

**CAUTION!**

Avoid damaging the top cover of the OLTC and the sealing surface of the on-load tap-changer head during disassembly and other operations. Make sure that the tap selector is completely immersed in transformer oil and the diverter switch compartment is filled with oil before operating the on-load tap-changer for the first time after the transformer has been dried. If the OLTC is dried inside the transformer tank, loosen the oil drain stopper. After drying, do not forget to remove the current switch insert, close the stopper, and then reinstall the insert.

**NOTE!**

Damaged OLTC O ring will cause oil leakage from oil compartment which will damage the OLTC.

8.1.2.2.1 Pass kerosene vapor into on-load tap-changer

The pipe joints R and Q on the tap-changer head (See Figure-44 for the position of the pipe joint on the tap-changer head) are connected to the connecting pipe, and the inner diameter is at least 50mm to connect to kerosene vapor.

8.1.2.2.2 Drying

- 1) Pass in kerosene vapor around 90°C and keep this temperature constant for 3 to 4 hours.
- 2) Increase the kerosene gas temperature at a rate of 10°C/hour till the specified final temperature, but the maximum temperature of the tap-changer cannot exceed 125°C.
- 3) In vacuum drying, the temperature on the tap-changer is 105°C to a maximum of 125°C. The highest residual pressure is 133Pa. The drying duration is the same as the transformer, but at least 50 hours.

8.1.2.2.3 To remove the diverter switch insert and close the oil drain stopper.

- 1) To remove the diverter switch insert according to section 6.5.1.
- 2) Close the oil drain stopper clockwise with an extended T-wrench. (Torque 20 N.m.)

8.1.2.2.4 After tightening the oil drain stopper, reinstall the on-load diverter switch insert according to 6.5.4.

8.2 Oil filling

After drying, in order to prevent too much moisture from entering the oil compartment, the oil compartment (diverter switch insert already installed) should be refilled with oil as soon as possible.

The top cover of the tap-changer should be closed again and 24 pieces of M10 bolts should be tightened, paying attention to the correct position of the sealing O-ring. Both the transformer and the diverter switch are filled with oil under vacuum. In order to vacuumize the diverter switch oil compartment and the transformer simultaneously, the by-pass pipe accessories provided by our factory can be installed between oil filling flange on the tap-changer head and oil-overflow flange of the transformer.

Fill the tap-changer with new transformer oil up to the same level as in the transformer tank. Thus, install the by-pass pipe accessories provided by HM between the oil filling flange of the tap-changer head and the oil-overflow flange of the transformer, so that the diverter switch oil compartment and the transformer can be vacuumized simultaneously. In order to meet the standard for oil filling of oil compartment and its conservator, use only new transformer insulating fluid in accordance with IEC 60296 for mineral oil, IEC 61099 for synthetic ester oil and IEC 62770 for natural ester oil. Both the on-load tap-changer and the transformer should be filled with new oil under vacuum at the same time.

- 1) In order to vacuumize the tap changer and the transformer at the same time, connect a by-pass pipe and the appropriate valves between pipe joints E2 and Q.
- 2) When filling oil, use the pipe joint S or Q on the tap-changer head. (Figure 42)
- 3) Do not dismantle the flexible by-pass pipes before shutting the connected valves.
- 4) For a better and precise evaluation of insulating fluid, measure the breakdown voltage (as per IEC 60156) and water content (as per IEC 60814) at a sample of $20^{\circ}\text{C} \pm 5^{\circ}\text{C}$ (as per IEC 60156) and do not forget to record temperature of the sample.
- 5) Check breakdown voltage and water content values specified at table-7.

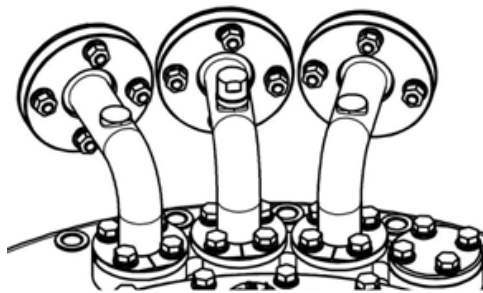


Figure 42 Pipe joint



NOTE!

Oil filled in the tap-changer's oil compartment and conservator must meet IEC 60296 and IEC 60422 standards.

**CAUTION!**

1. After drying and without oil, continuing operation will cause damage to the tap-changer!
2. After the transformer is dried, before operating the on-load tap-changer for the first time, make sure that the tap selector is completely immersed in the transformer oil and the diverter switch oil compartment is filled with oil.
3. Do not step on the rupture disc when filling oil, and cannot directly fill oil from the hole of the rupture disc. (Figure 43, Figure 44)

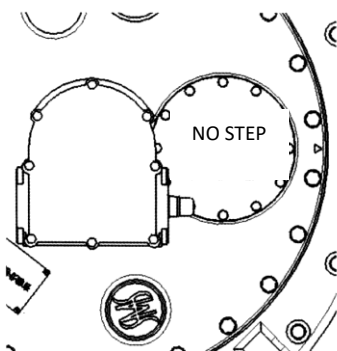


Figure 43 Do not step on rupture disc

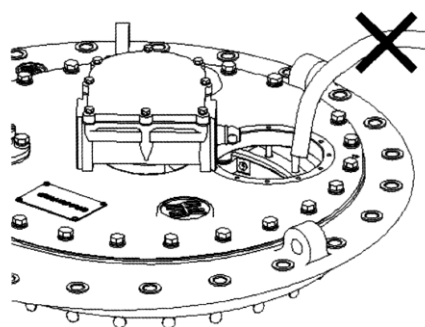


Figure 44 Do not fill in oil through rupture disc

9. Installation of tap-changer protection device and transmission device components

9.1 Installation of gas relay

Install according to the instruction manual provided when the gas relay is supplied.

9.2 Use of pressure relief device

The red bar on the top of the pressure relief device must be removed before operation (Figure 45).

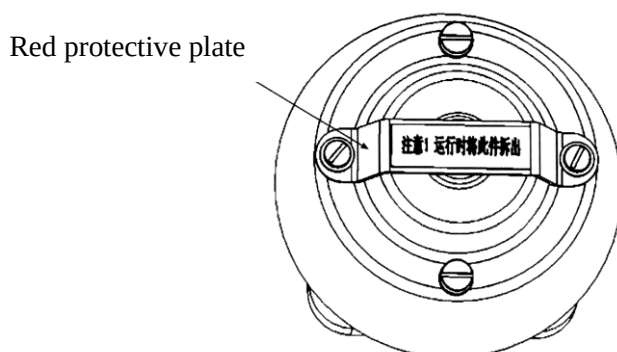


Figure 45 Pressure relief device

9.3 Installation of motor drive unit

Install according to the MDU instruction manual provided by HM.

9.4 Installation of bevel gear box

The bevel gear box is to be mounted onto the supporting bracket of the transformer tank by 2 bolts with 18mm bore diameter.



NOTE!

The serial number marked on the nameplate of the bevel gear box must be consistent with the serial number on the nameplate of the tap-changer.

When horizontal and vertical drive shafts are over two meters, an intermediate support gear box must be added. The installation of the specially designed bevel gear box, steering gear box and the intermediate support gear box of the vertical or horizontal drive shaft can be carried out according to the above instructions.

The gear box is fixed on the mounting flange with 3 pressing plates (each board with 2 bolts), and the angle can be adjusted according to user requirements.

- 1) Loosen the pressing plate of the gear box and pull out one end of the board slightly (6 pieces of M8 bolts, 13mm wrench). (Figure 46)
- 2) Turn the gear box to the required installation position by rotating the drive shaft of the gear box. (Figure 47)
- 3) Tighten the gear box pressure plate, and tighten the bolt (Torque 15 N.m.).



CAUTION!

1. The gear box mustn't rotate on its own.
2. When rotating the gear box, pay attention to avoid bumping the pressure relief valve (if existing.)



CAUTION!

Incorrect adjustment of the gear box can cause damage to the tap-changer! The gear box can only be adjusted when the pressure ring is loosened. It must not rotate when it is adjusted. Only turn the drive shaft to adjust the gear box.

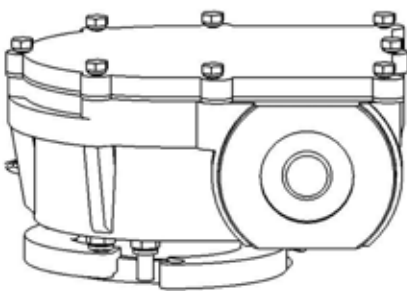


Figure 46 Gear box

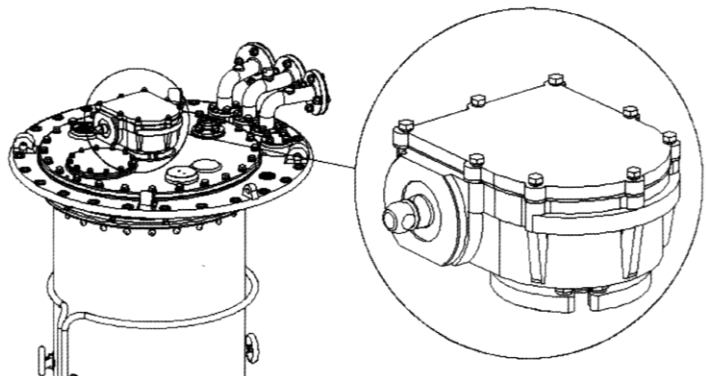


Figure 47 Turn the gear box

9.5 Installation of horizontal and vertical drive shafts

- 1) Install the horizontal and vertical drive shafts according to the relevant Huaming drive shaft instruction manual. (Figure 48)
- 2) Connect the drive shaft with motor drive unit and tap-changer according to the instruction manual of Huaming motor drive unit. (Figure 49)

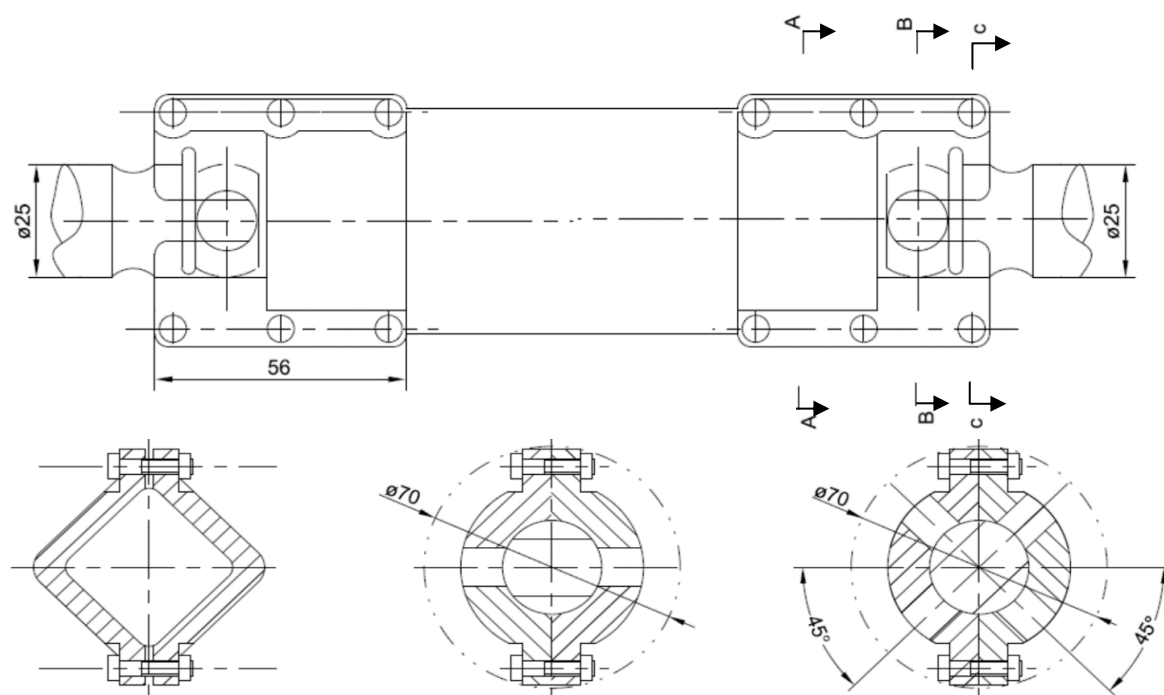


Figure 48 Install the drive shaft

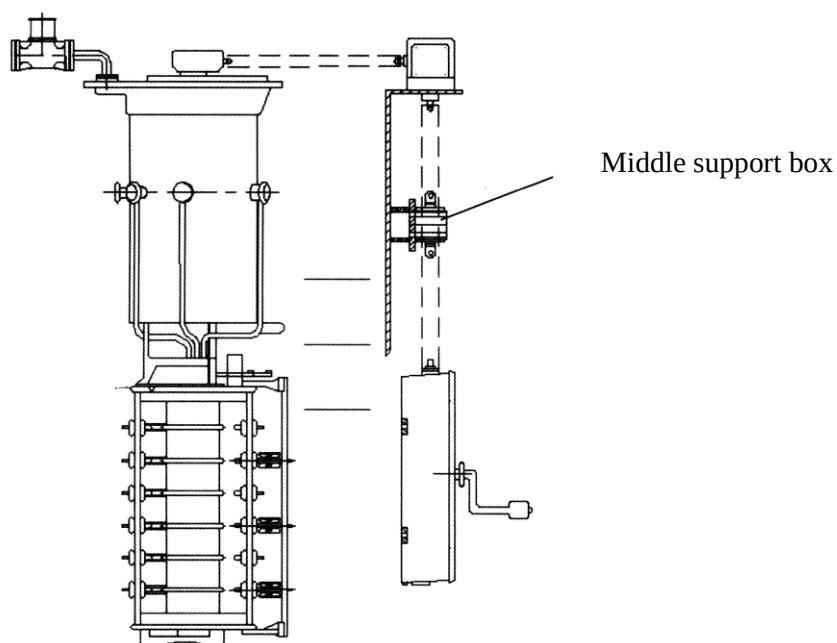


Figure 49 Drive shaft connection

**CAUTION!**

Incorrect adjustment of the gear box can cause damage to the tap-changer! The gear box can only be adjusted when the pressure ring is loosened. It must not rotate when it is adjusted. Only turn the drive shaft to adjust the gear box.

**NOTE!**

When the length of the vertical drive shaft exceeds 2m, a middle support box is required

9.6 Installation of driving shaft shield cover

9.6.1 Installation of horizontal shield cover

The length of the horizontal shield is determined by the design of the transformer manufacturer. After installation of the driving shaft, put the clamps on both ends of the horizontal shield covers (provided in the accessories), then clamp them on the protruding platform of the gear box and the bevel gear box, and finally move the clamps to the protruding platform positions on both sides, tighten and firm. (Figure-50)

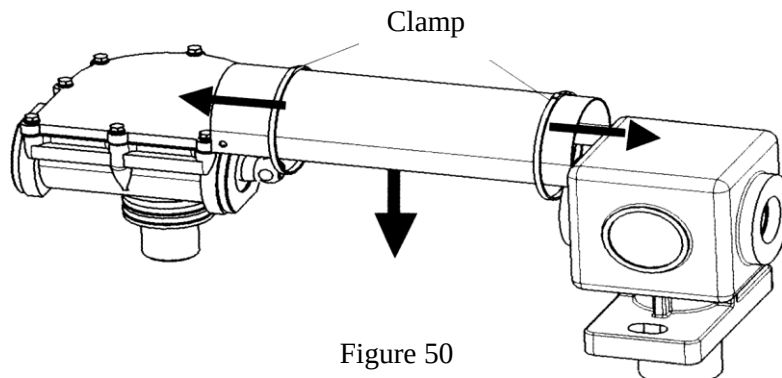
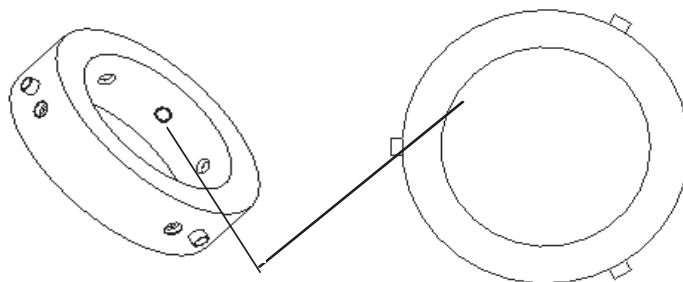


Figure 50

9.6.2 Installation of vertical shield cover

9.6.2.1 Fix 3 stainless steel 6×16 straight pins on the bushing (120° equally divided), and protrude the outer diameter of the bushing about 5mm, and the inner diameter positions should not be protruded. (Figure 51)

9.6.2.2 Place the bushing at the lower end of the bevel gear box and fix with 4 pieces of M6×10 set screws (90° equally divided). (Figure 52)



Straight pins not protruded
the inner diameter

Figure 51 Fix steel

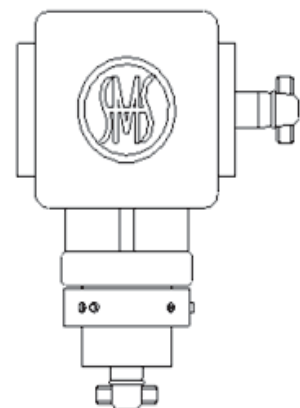


Figure 52 Place bushing

9.6.2.3 Put the two shield covers outside of the driving shaft with the gaps at both ends, and the thicker one on the top. Then connect it to the middle of the bevel gear box and motor drive unit according to the figures below. (Figures 53 & 54)

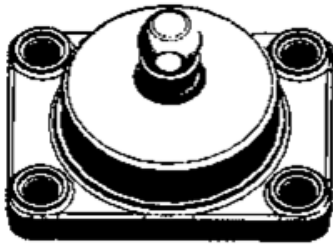


Figure 53 CMA7 connection to the drive shaft



Figure 54 Connect to the bevel gear box



Figure 55 Place the lower shield

9.6.2.4 Place the lower shield cover outside of the bushing gaps to avoid 4 bolts. (Figure 55)

9.6.2.5 Pull up the protective cover to the lower end of the square gear box, and set it on the bushing, with the 3 notches facing directly, 3 cylindrical pins, and then rotate the protective cover so that the cylindrical pins are just stuck in the horizontal groove (Figure 56). (3 pins are only for temporary hang)

9.6.2.6 At this time, the two protective covers have been positioned, use two clamps on the thinner protective cover at the bottom. Clamp the thicker protective cover on the upper part (Figure 57)

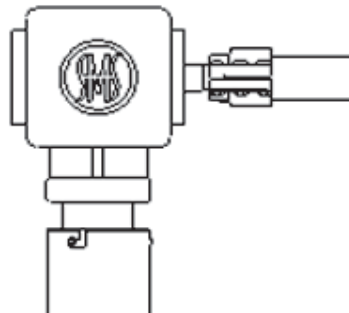
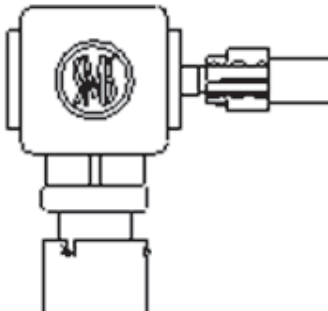


Figure 56 Rotate the protective cover



Figure 57 Clamp the protective cover

9.7 Installation of tap-changer group

The arrangement of tap-changer group composed of several units of tap-changers is quite special, and every tap-changer is at the setting position. The connection steps are as follows:

- 1) Check whether the operating positions of all on-load tap-changers are the same (check from the tap-changer head observation window and every tap-changer must be at the adjusting position).
- 2) Loosen the pressing ring of the gear box and slightly pull out one end of the ring (6 pieces of M8 bolts, 13mm wrench).
- 3) Turn the gear box to the required installation position by rotating the drive shaft of the gear box.
- 4) Tighten the pressing ring of the gear box, and lock the bolt with a locking tab (Torque 15 N.m.).
- 5) Pay attention to the serial number printed on the flange of the gear box and the arrow below. The direction of the arrow indicates the steering direction of the drive shaft when using a hand crank to clockwise rotate the MDU. The direction of all arrows on all gear boxes must be the same. The gear box can be adjusted only when the pressing ring is loosened.



CAUTION!

The gear box can only be adjusted only by rotating its drive shaft. It must not rotate on its own.



CAUTION!

Incorrect adjustment of the gear box can cause damage to the tap-changer!

- 6) Rotate each shaft end counterclockwise to run every step of the tap-changer once, until the diverter switch operates for one full cycle.
- 7) Check whether the operating positions of every tap-changer head and the MDU are the same.
- 8) Install horizontal drive shafts between every tap-changer head. Each tap-changer has its own shaft coupling. Start with the tap-changer closest to the MDU.
- 9) After all the drive shafts are installed, the drive shaft of the gear box should continue to rotate for 2.5 cycles counterclockwise to correctly complete the tap change operation.
- 10) Rotate the drive shaft clockwise to adjust the tap-changer back to the calibration position. Then operate together with the diverter switch. The drive shaft of the gear box should continue to rotate for 2.5 cycles in the clockwise direction to make the tap change operation completed correctly.
- 11) Make sure that all tap-changers are operate (a clear sound can be heard) at the same time.
- 12) Check whether the operating positions of every tap-changer head and the MDU are the same.
- 13) Install the vertical drive shafts.

9.8 Tap position calibration of tap-changer transmission system

When the tap-changer is connected to the MDU, manual operation for one cycle before electric operation is required

When OLTC coupling with MDU, the time interval between the moment the diverter switch operates and the the completion of MDU operation is required to be the same for the two rotation directions.

The connection verification of the tap-changer and the MDU normally is performed before FAT test. But in order to ensure the reliability of the tap-changer, below procedure shall be applied for the connection verification.

- 1) Rotate in the $1 \rightarrow N$ direction. When the diverter switch operates (a clear sound will be heard), continue to rotate the hand crank and record the number of rotations, until the red mark in the green band on the MDU tap change operation indicator wheel appears in the middle of the observation window. Stop rotating and write down the number of rotations as “m”.
- 2) Rotate the hand crank in the opposite direction $N \rightarrow 1$ to return to the original setting position, and record the number of rotations as “K” (follow instruction step a)
- 3) If the number of rotations $m=k$, the connection is correct. If $m \neq k$ and $m-k > 1$, balance of the rotation difference is required. Loosen the connection of MDU’s vertical shaft, rotate $(m-k) / 2$ laps in the direction which has more rotations, and finally connect the vertical drive shaft with the MDU.
- 4) Following the above steps, check the rotation difference between the MDU and the tap-changer, until the correct number of rotations, that is, $m=k$.

For example:

The connection verification between the CM tap-changer and the SHM MDU: from position 10 (setting position) to position 11, $m=5$ rotations; from position 11 back to 10 (the original setting position), $k=3$ rotations. The hand crank rotation difference: $m-k=5-3=2$. Adjust the number of rotations $(m-k) / 2 = (5-3) / 2 = 1$ rotation. Loosen the connection between the vertical drive shaft and the MDU, and turn the hand crank for one circle in the direction of position 10 to 11 according to the number of rotations calculated above. Then connect again. Check whether the rotation difference between the two directions is balanced:

- 1) Record the rotation numbers of m and k when connected
- 2) Disconnect the connection and rotate for $(m-k)/2$ circles in the direction with more rotations
- 3) Reconnect and verify until $m=k$

10. Tap-changer Factory test and test preparation

10.1 Test preparation

10.1.1 OLTC Fully gas release

Before putting into operation for the first time, use the gas discharge stopper in the oil suction pipe on the tap-changer head to deflate the connection pipe.

10.1.1.1 Gas release

- 1) Remove the M30 bolt cap on the bleeding device E1 (36mm wrench).
- 2) Pry up the valve stem with a screwdriver to deflate the tap-changer head (Figure 58)
- 3) Tighten the bolt cap on the bleeding device E1 (Torque 9 to 12 N.m.)

10.1.1.2 Gas release for S: connecting flange for suction pipe

- 1) Remove the M16 bolt cap on the pipe joint S (16mm wrench) (Figure 59).
- 2) Open the notched gas discharge stopper M6 to release gas.
- 3) Close the stopper (Torque 2 N.m.).
- 4) Close the bolt cap on the stopper. (Torque 8 to 10 N.m.)

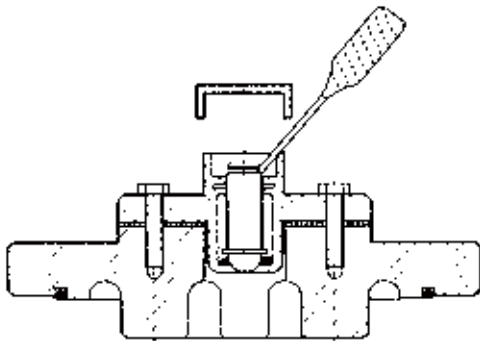


Figure 58 Pry up the valve stem

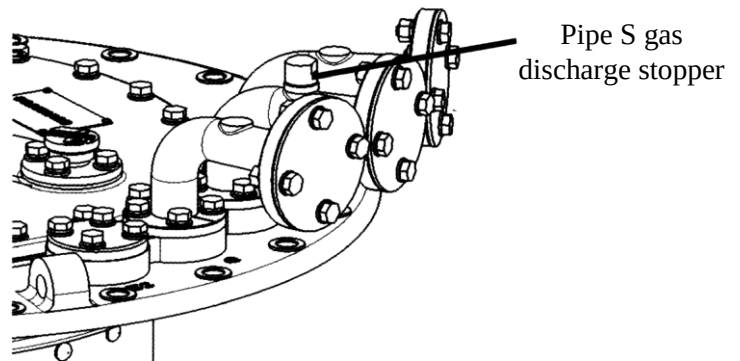


Figure 59 Remove the bolt cap

10.1.2 Grounding

- 1) Connect the grounding bolt of the tap-changer head to the transformer tank cover (1 piece of M12 bolt and nut) with a 19mm wrench, torque of 50 to 60 N.m. (Figure 60).
- 2) Connect the transmission mechanism's grounding bolt M12 (19mm wrench, torque 50 to 60 N.m.) connected to the transformer tank.

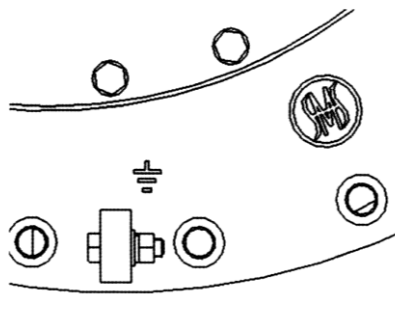


Figure 60 Connect the grounding bolt

10.2 Test in transformer factory

10.2.1 Operation test

Pre- operation must be carried out to check the mechanical functions of the tap-changer and the motor-drive mechanism before the transformer is energized.

**CAUTION!**

Continue to operate the tap-changer **WITHOUT OIL** will cause damage to the tap-changer !

When operating the tap-changer for the first time after the transformer is dried, make sure:

- 1) The tap selector is completely immersed in transformer oil and the diverter switch oil compartment is filled with oil
- 2) During the operation test, the tap-changer must run the complete operating range.

**NOTE!**

Make sure MDU's and tap-changer's position indicators show the same position.

**CAUTION!**

If the connection between the tap-changer and the MDU is improperly aligned, continuous operation will cause damage to the tap-changer.

- 3) Make sure that the MDU's and tap-changer's position indicators (observation window on the tap-changer head) show the same position in every operating position.
- 4) Check the electrical and mechanical end limit functions at two end positions (Please refer to MDU instruction manual).

10.2.2 The electrical test of the transformer

**DANGER!**

There is explosive gas under the tap-changer cover, in the connection system, oil conservator and the outlet of air dryer. The flying parts and the high temperature oil splashing to the outside will cause fatal and serious injury.

**NOTE!**

Please confirm: there is no open flame, heat source or electric spark (such as electric spark produced by static electricity) within its direct contact range, and there is no possibility of generating them. It must be ensured that the tap-changer test must be carried out by professionals in accordance with the test procedures.

11. Transformer transportation and put into operation on site

11.1 Remove the MDU for transportation

- 1) Place the MDU at the calibration position and then remove the coupling
- 2) Remove the MDU
- 3) Do not operate the MDU when the tap-changer is not connected well. The re-assembly of the MDU and drive shaft must be carried out according to Chapter 9.3 to 9.7 of this instruction manual.

11.2 Oil filling and transportation without oil conservator

If the transformer is filled with oil during storage and transportation but no oil conservator is installed, a bypass pipe must be installed between the inside of the diverter switch oil chamber and the transformer oil tank to balance the pressure generated by oil expansion. The bypass tube is installed between the pipe connector E2 and Q.



NOTE!

For the storage period of 3 to 4 weeks without oil conservator, the oil level of the tap changer should be lowered by about 5 liters.

11.3 Transportation without oil

If the transformer is transported or stored without oil filling, all the oil in the tap-changer oil compartment must be discharged. The protection method for tap-changer oil compartment should be same as that for the transformer, such as nitrogen filling.



NOTE!

For long-term storage, the heater of the MDU should be regularly energized to prevent moisture.

11.4 Putting tran



DANGER!

There is explosive gas under the tap-changer cover, in the connection system, oil conservator and the outlet of air dryer. The flying parts and the high temperature oil splashing to the outside will cause fatal and serious injury.



NOTE!

Please confirm: there is no open flame, heat source or electric spark (such as electric spark produced by static electricity) within its direct contact range, and there is no possibility of generating them. It must be ensured that the tap-changer test must be carried out by professionals in accordance with the test procedures.

Before putting the transformer into operation, make sure the tap selector is completely immersed in the transformer oil and the oil compartment is filled with oil.

- 1) Fill OLTC with oil according to Chapter 8.2 (Refer to Chapter 13.1 Table 7 for the limit of oil quality).



NOTE!

Please confirm: there is no open flame, heat source or electric spark (such as electric spark produced by static electricity) within its direct contact range, and there is no possibility of generating it.

- 2) To fully release gas in OLTC, please follow Chapter 10.1.1
- 3) For pre-operation procedure, please follow Chapter 10.2.1
- 4) Please make sure that the OLTC oil conservator's lowest oil level signal contact is connected to the circuit breaker's trip circuit.
- 5) Check whether the gas relay function is normal according to the Huaming gas relay QJ-25 series instruction manual.
- 6) The protective strip of the pressure relief valve must be removed before putting it into operation. Follow safety instructions of all functional inspections and tests during commissioning and the safety instructions described in Chapters 9.2 and 10.2.2.

12. Tap-changer operation monitoring and maintenance

**CAUTION**

Inspection of OLTC head cover, protective relay and MDU should be carried out regularly.

- Whether there is leakage at the connection between the protection relay and the pipeline;
- The condition of the MDU sealing and all electric components;
- If the protective relay alarms, a thorough inspection should be done to the transformer and OLTC. If necessary, the tap-changer body can be lifted out for inspection.
- Before putting into operation again, OLTC and transformer must be inspected for damages.

**NOTE!**

Please contact Huaming service team if OLTC and MDU have severe problems or the protective relay actions and it's not easy to fix on site.

13. Maintenance

Danger!



Electric shock!

An energized transformer may cause death or serious injury.

- ▶ Turn off the high voltage and low voltage ends of the transformer.
- ▶ Lock the transformer to prevent accidental restart.
- ▶ Ensure that all components are powered off.
- ▶ Clearly ground all transformer terminals (ground wire, ground isolation switch) and short-circuit them.
- ▶ Cover or isolate adjacent live parts.

Danger!



Electric shock!

Operation of energized on-load tap-changers may cause personnel death or serious injury.

- ▶ Disconnect all auxiliary circuits, such as tap change monitoring devices, pressure relief valves and pressure monitors.
- ▶ Ensure that all components are powered off.

Warning!

Risk of explosion!

Explosive gas in the oil compartment of the on-load tap-changer, transformer, piping system, oil conservator, and the outlet of the moisture absorber may ignite or explode, causing serious injury or death.

- ▶ Ensure that there are no open flames, hot surfaces, or electric sparks (for example, electric sparks caused by static electricity accumulation) in the environment around the transformer, and that the above situations will not occur.
- ▶ Do not operate any electrical equipment (for example, impact wrenches have the risk of sparking).
- ▶ Only use conductive and grounded hoses, pipes and pump operation equipment suitable for flammable liquids.

Warning!



Check the motor drive unit!

Condensation in the MDU protection case can cause damage to MDU.

- ▶ Always keep MDU's protection case sealed.
- ▶ If the operation is interrupted for more than 2 weeks, connect and operate the anti-condensation heater in the MDU. If this requirement cannot be met (for example, the heater cannot be energized during transportation), a sufficient amount of desiccant must be placed inside the case.

Remind

13.1 Inspection

Please follow the inspection instruction below:

Interval	Instruction				
Once per year	Check the door tightness, cable bushings, MDU protective cabinet and control panel's ventilation device				
Once per year	Check the tightness of tap-changer head cover, protective relay and all pipes' connectors.				
Once per year	Check the MDU's protective cabinet and if the heater inside the control panel works normally or not.				
Once per year	Check if the protective relay functions normally.				
Once per year	Check if the Silica gel desiccant inside the OLTC oil conservator work normally				
Once per year	Check the oil sample. If it doesn't meet the parameters below, it's recommended to change the transformer oil.				
	OLTC equipment highest voltage (kV)	Oil before operation		Oil during operation	
		Water (mg/L)	Breakdown voltage (kV/2.5mm electrode)	Water (mg/L)	Breakdown voltage (kV/2.5mm electrode)
	300	≤15	≥50	≤25	≥40
	252	≤15	≥50	≤25	≥40
	170	≤20	≥45	≤30	≥35
	126	≤20	≥45	≤30	≥35
	72.5	≤20	≥35	≤40	≥30

Table 7 Inspection instruction

13.2 Maintenance interval

Interval	Instruction
After 200,000 times or 2-year OLTC operation (based on the number shown in MDU's counter) whichever comes first	Lift out the switch insert for inspection. Regarding this, please contact Huaming's after-sales department.
After the 1 st inspection, every 7 years or after the number of operations indicated in Table 9 (based on the number shown in MDU's counter) whichever comes first	Routine maintenance. Regarding this, please contact Huaming's after-sales department.
After 250,000 times OLTC operation (based on the number shown in MDU's counter)	Replace arcing contacts' braided contact leads even when they are not damaged and when contacts do not need to be replaced. Regarding this, please contact Huaming's after-sales department.
After 800,000 times OLTC operation (based on the number shown in MDU's counter)	Replace the diverter switch. Regarding this, please contact Huaming's after-sales department.

Note:

1. If OLTC does not operate or energized for a long time, at least make five full cycles of operations after the power is cut off.
2. Tap selector usually does not need maintenance unless when used in industrial transformers, where maintenance is needed after 1,000,000 operations.

Table 8 Maintenance interval

OLTC	Transformer's rated current	OLTC Operation Times	
		Without online oil filter	With online oil filter
CM III 500	$\leq 400A$	100,000	150,000
	$> 400A$	80,000	150,000
CM III 600	$\leq 400A$	100,000	150,000
	$> 400A$	80,000	150,000
CM I 500	$\leq 500A$	100,000	150,000
CM I 600	$\leq 600A$	100,000	150,000
CM I 800	$\leq 500A$	100,000	150,000
	$> 500A$	80,000	150,000
CM I 1200	$\leq 800A$	80,000	150,000
	$> 800A$	70,000	140,000
CM I 1500	$\leq 1200A$	70,000	140,000
	$> 1200A$	60,000	120,000

Table 9 OLTC operation times

Please write to export@huaming.com for any need of support.

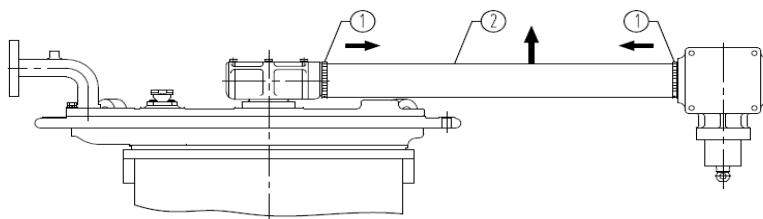
13.3 Change tap-changer transformer oil

13.3.1 Record the current position of tap-changer

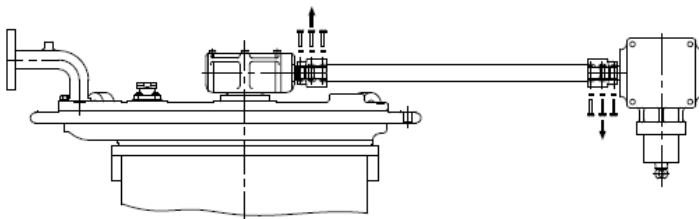
13.3.2 Switch tap-changer to the setting position (which indicated in connection diagram)

13.3.3 Disassembly of horizontal drive shaft

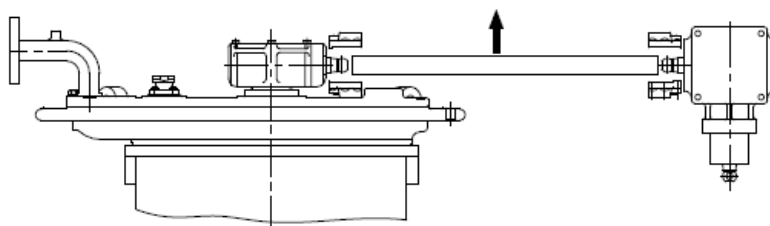
- If the MDU operates when it is not coupled, the OLTC and transformer will be damaged!
 - If you remove the horizontal drive shaft, do not operate the MDU!
- a. Remove the 2 clamps on the protective cover of the horizontal drive shaft, and then remove the protective cover.



b. Unscrew the 6 bolts on the coupling between the gear box and the bevel gearbox

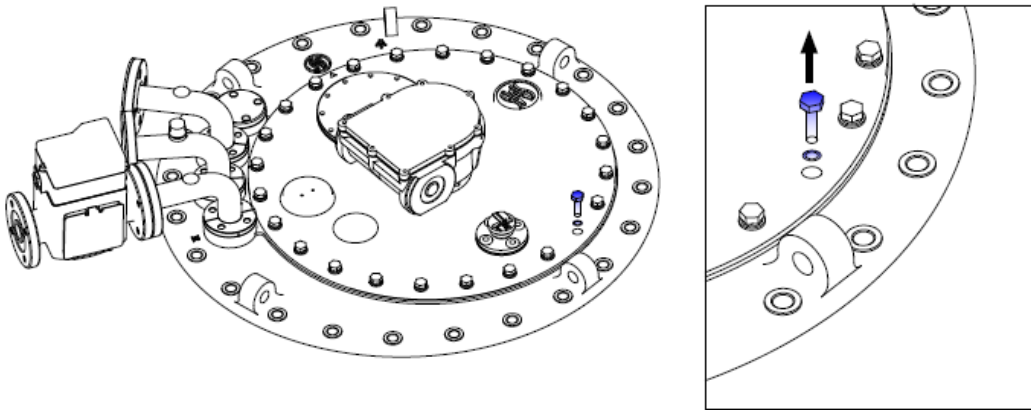


c. Remove the horizontal drive shaft. Ensure that the shaft pin is not lost

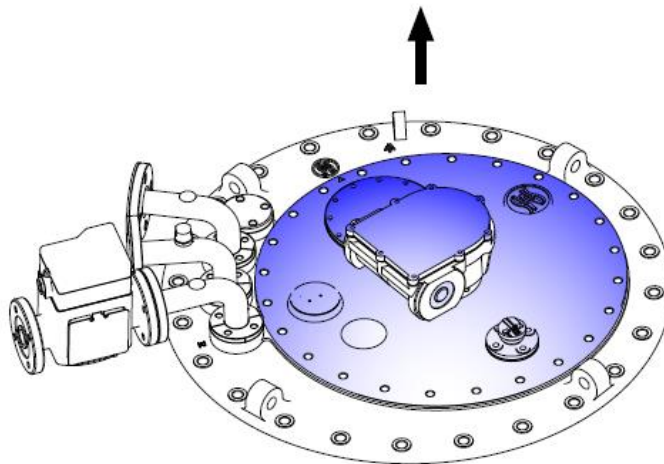


13.3.4 Drain the oil compartment and oil conservator

- 1) Make sure to open the shut-off valve between the oil conservator and the OLTC.
- 2) Remove the screw cap on bleeding E1 on the OLTC head cover.
- 3) Use a screwdriver to pry up the stem of E1. Then, gas will escape under the tap-changer head cover.
- 4) After the gas is exhausted and the transformer oil flows out of the bleed valve, close the bleed valve.
- 5) Close the shut-off valve between the oil conservator and the OLTC.
- 6) Open E1 again, and draw out about 10 liters of transformer oil through the pipe joint S until there is no transformer oil in the area under the head cover of the OLTC.
- 7) Remove the screws and washers on the OLTC's head cover.



- 8) Remove OLTC's head cover



- 9) Pump out the transformer oil through the pipe joint S.
- 10) open the shut-off valve between the oil conservator and the OLTC. The transformer oil in the oil conservator will flow into the oil compartment.
- 11) Pump out the transformer oil through the pipe S.

13.3.5 Drying and oil filling refer to Chapter 8

13.4 Notice

**CAUTION**

OLTC must be maintained strictly follow the manual to ensure its good performance. Otherwise, it will cause damages to operations of tap-changer and transformer.

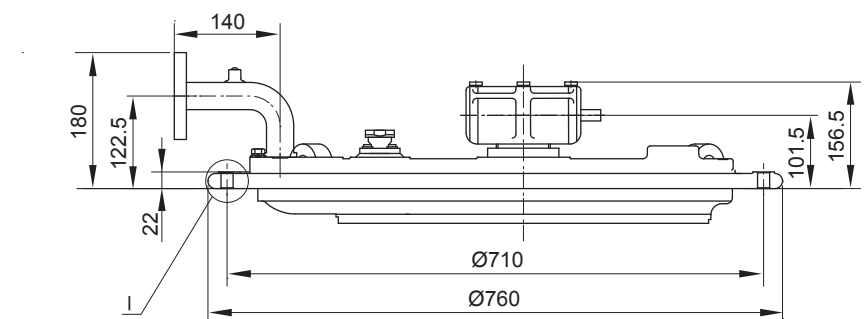
**CAUTION**

If the tap-changer does not regulate the voltage for a long time or has a tap position that is not used for a long time, no less than 5 cycles of tap change operation should be performed in the full tap range during a power cut.

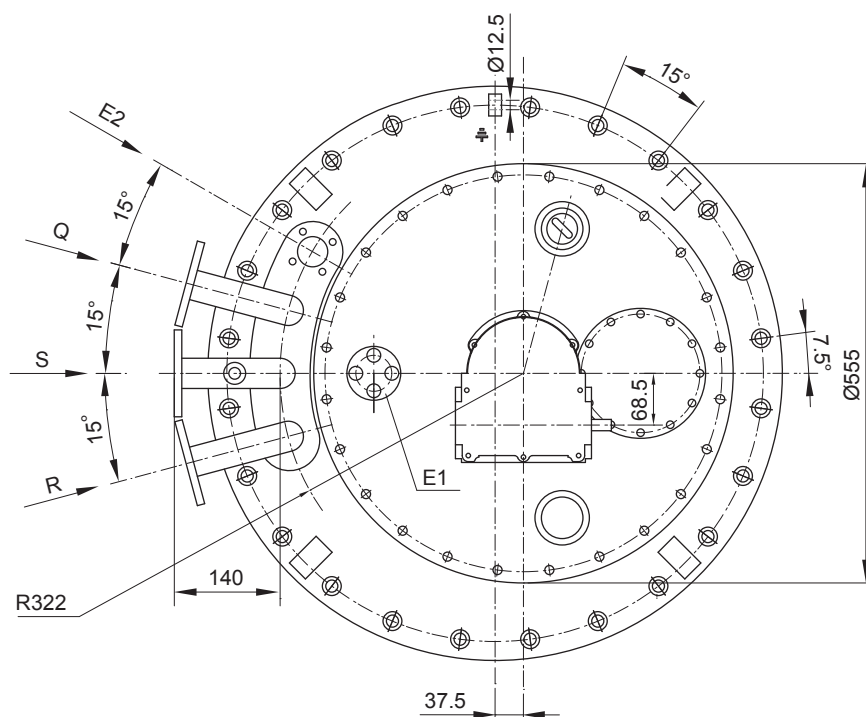
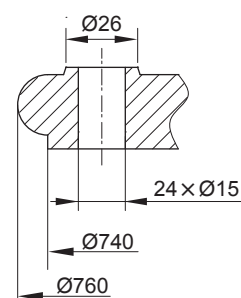
14. Appendix

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Appendix 1 CM OLTC mounting flange dimension

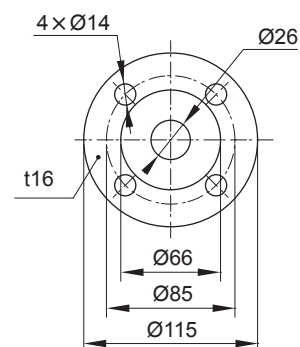


I: Magnified



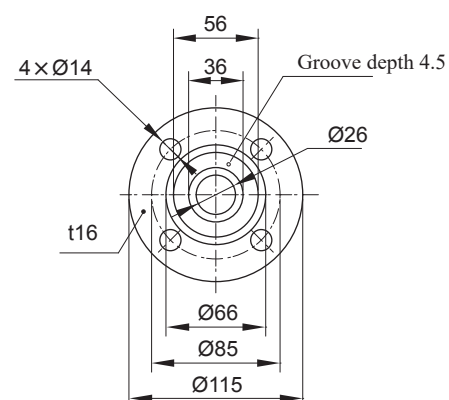
Direction R

Connecting flange



Direction S Q (with groove)

Connecting flange



Unit: mm

E1: Bleeding for OLTC

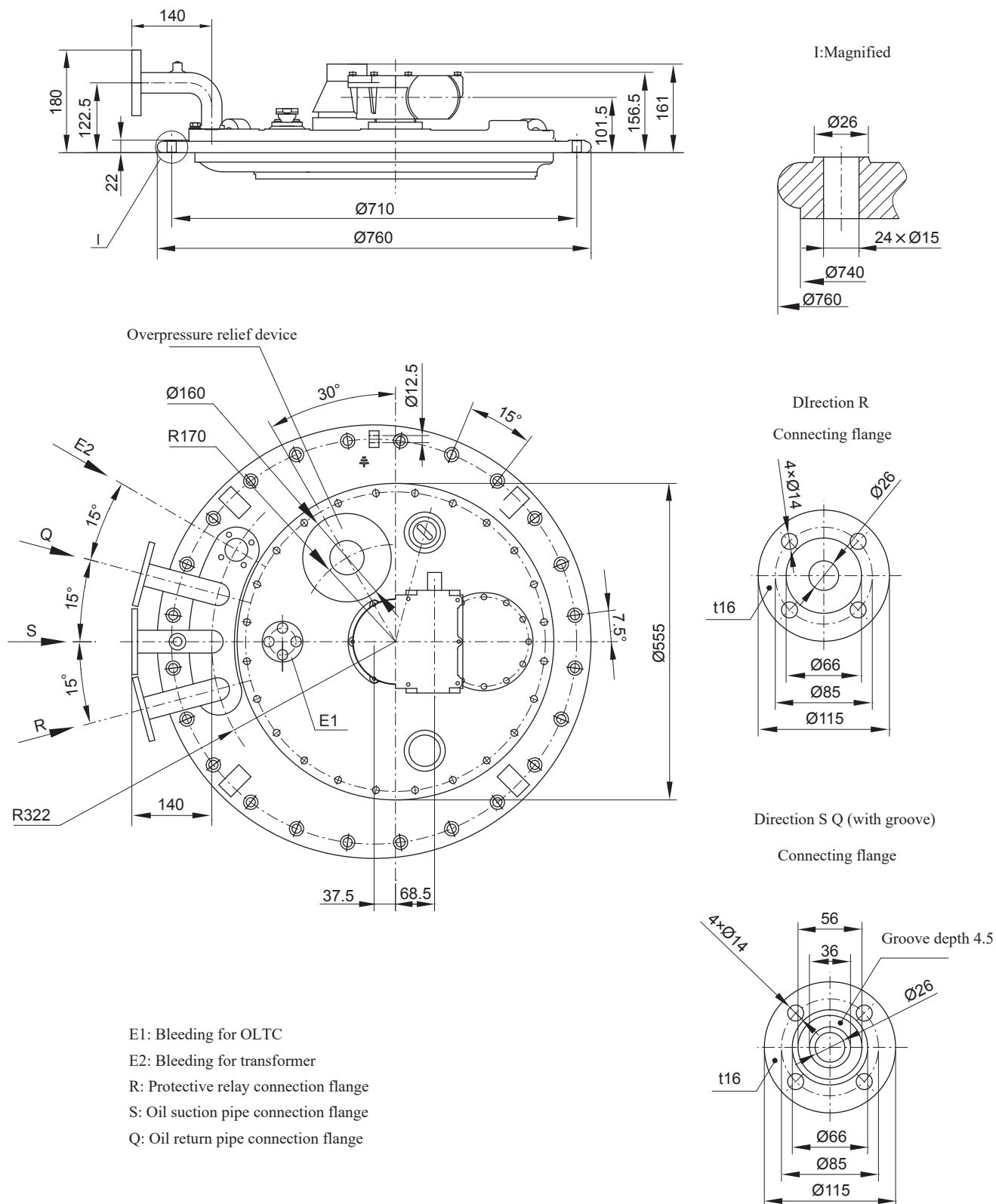
E2: Bleeding for transformer

R: Protective relay connection flange

S: Oil suction pipe connection flange

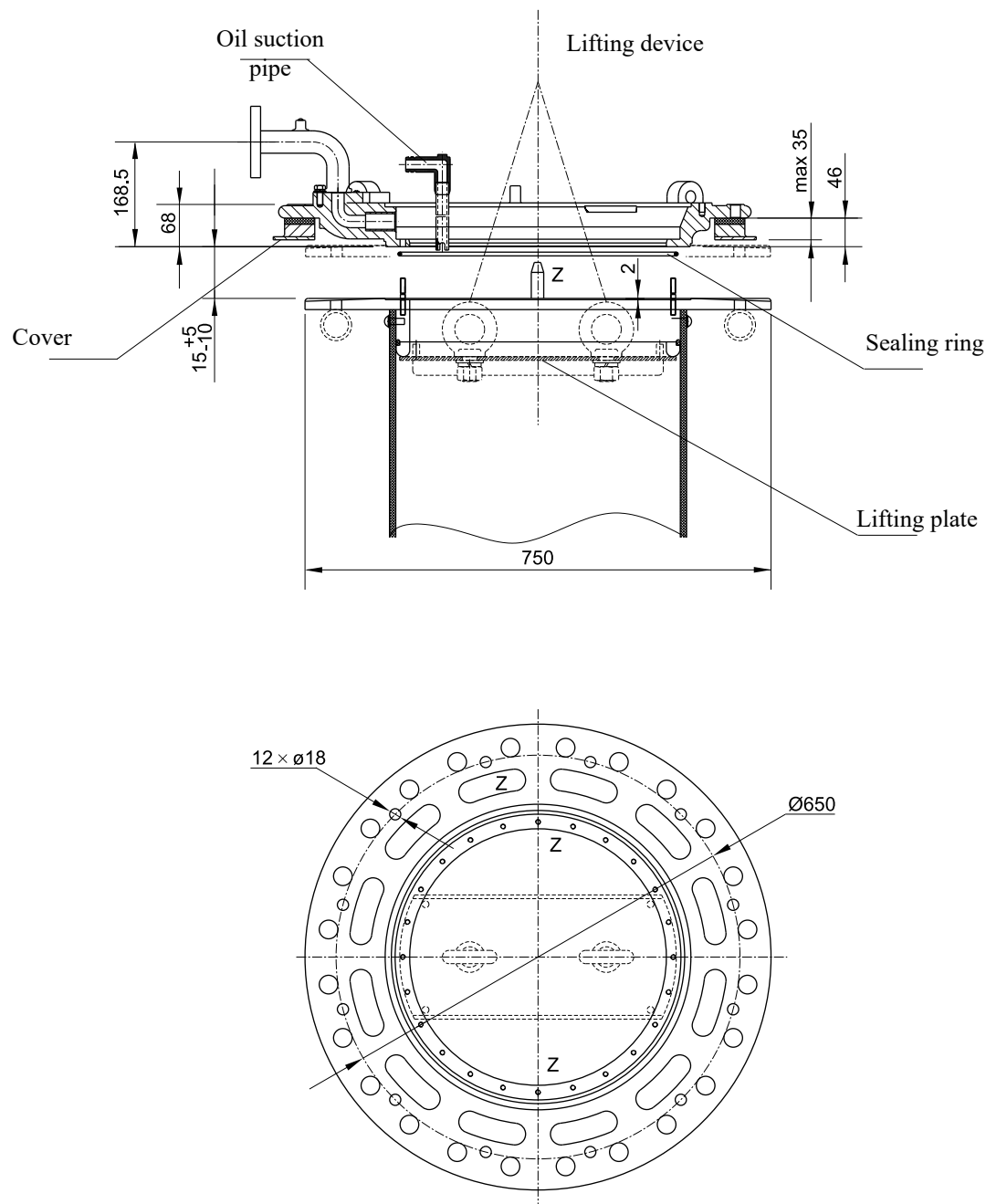
Q: Oil return pipe connection flange

Appendix 2 CM OLTC mounting flange with pressure relief device dimension



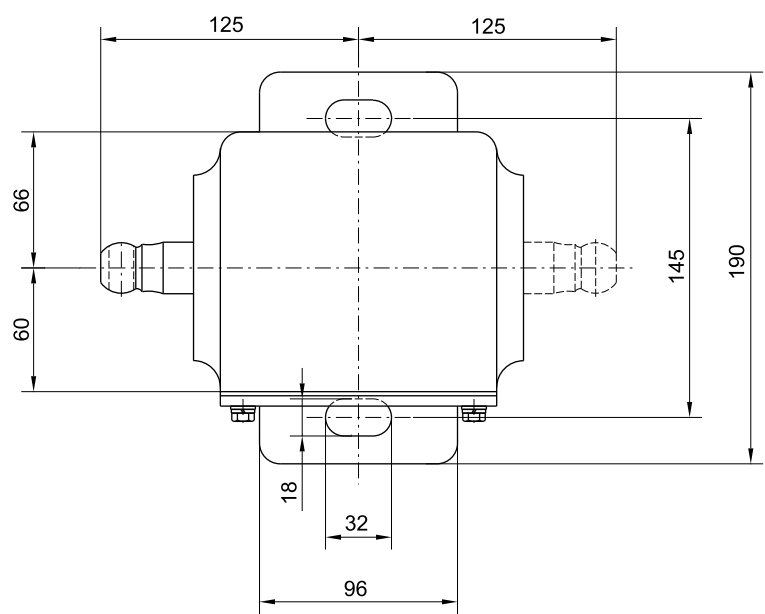
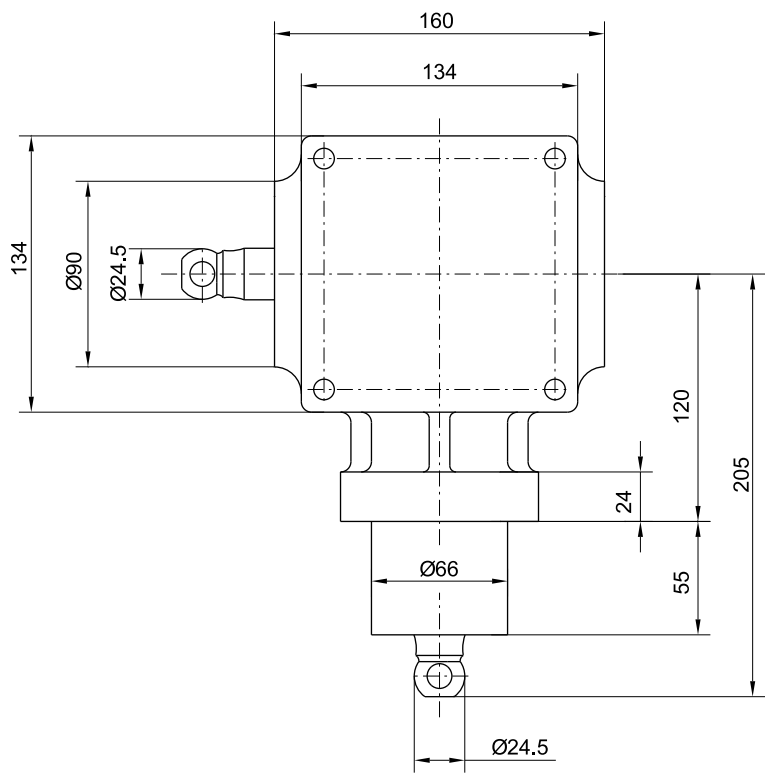
Unit: mm

Appendix 3 CM OLTC mounting flange dimension for bell-type tank



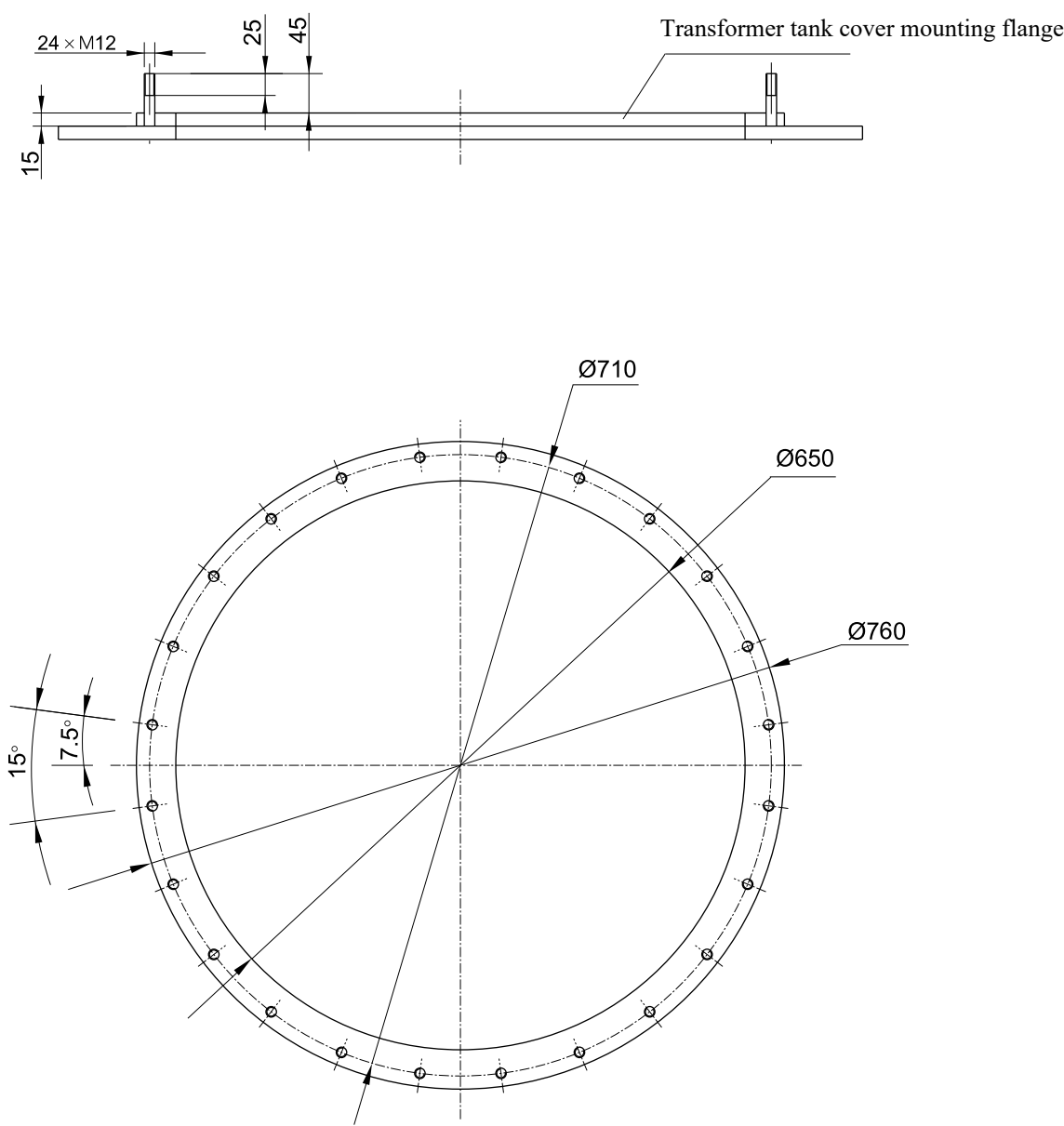
Unit: mm

Appendix 4 Bevel gear box dimension



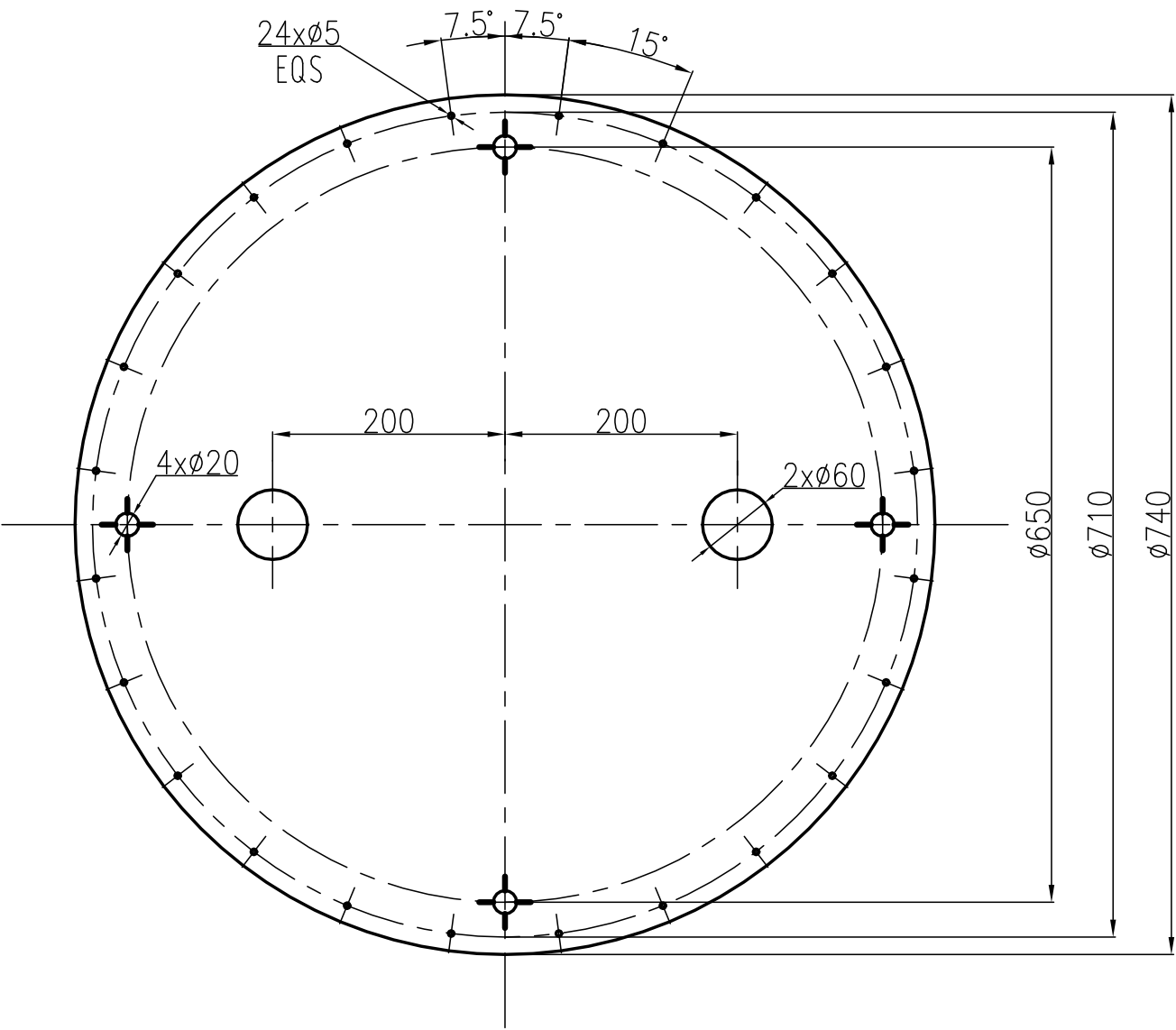
Unit: mm

Appendix 5 Transformer connection flange for CM OLTC



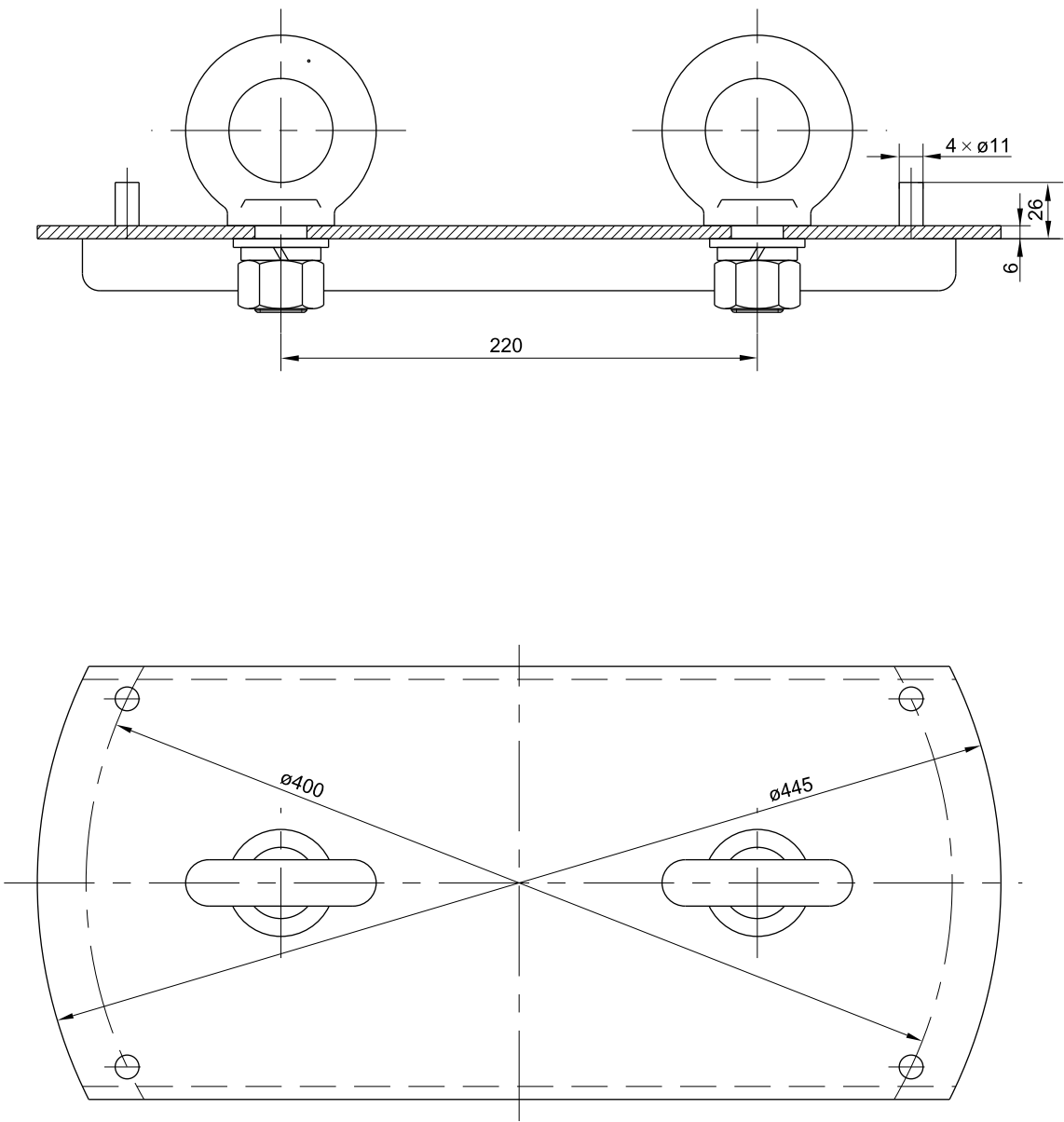
Unit: mm

Appendix 6 Drilling template for CM OLTC



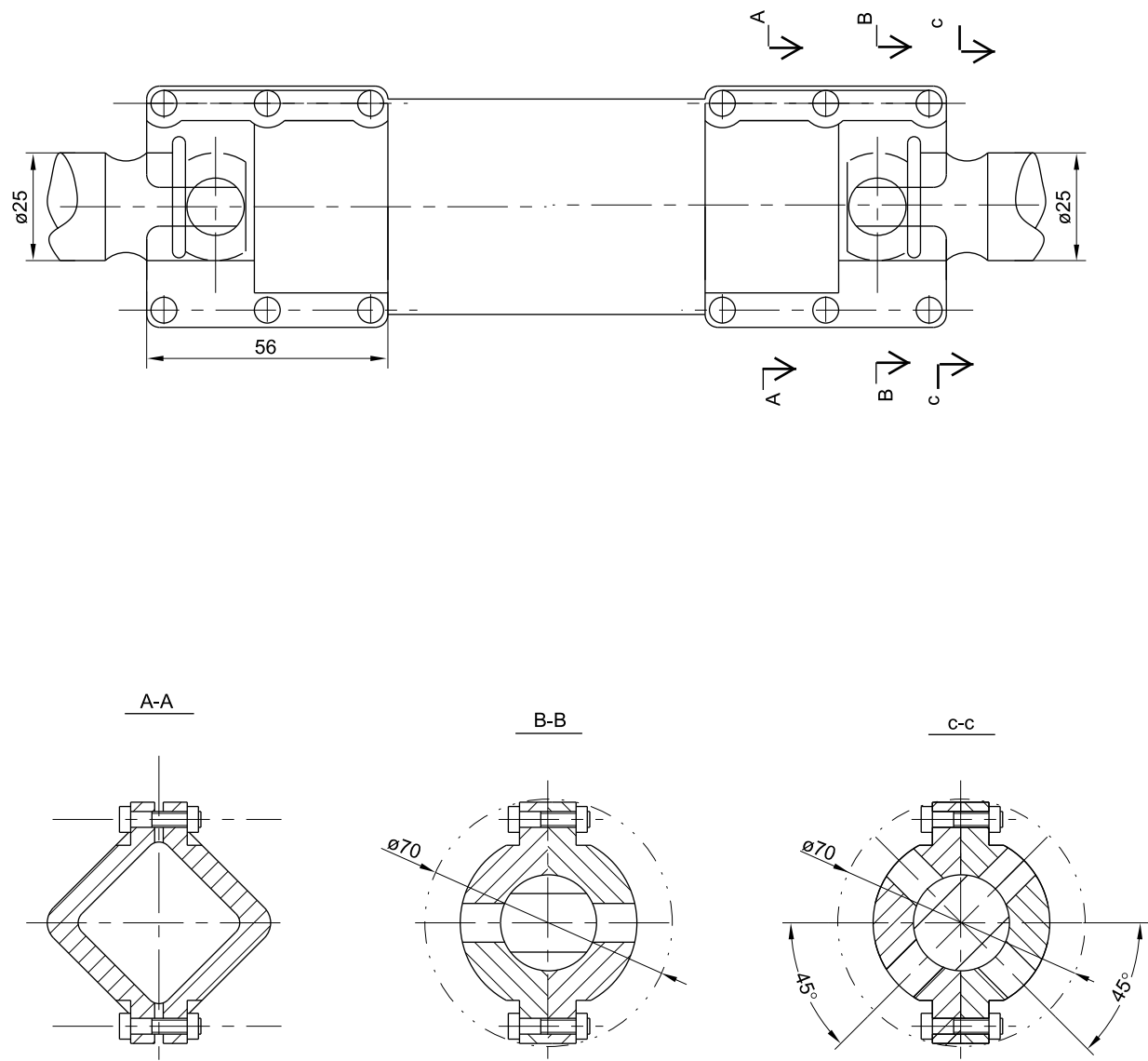
Unit: mm

Appendix 7 Lifting plate dimension for bell-type tank



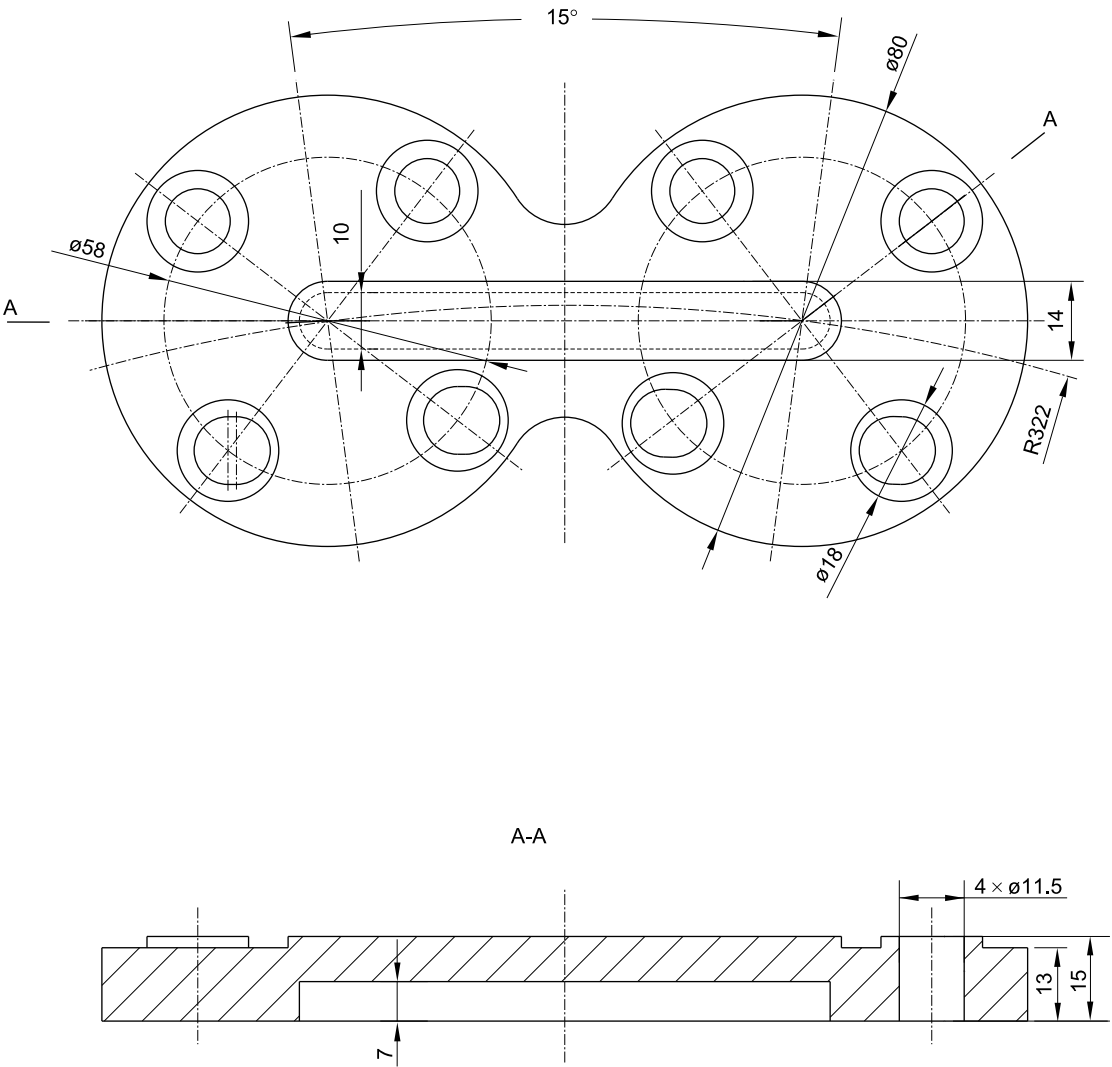
Unit: mm

Appendix 8 Installation of vertical and horizontal drive shafts



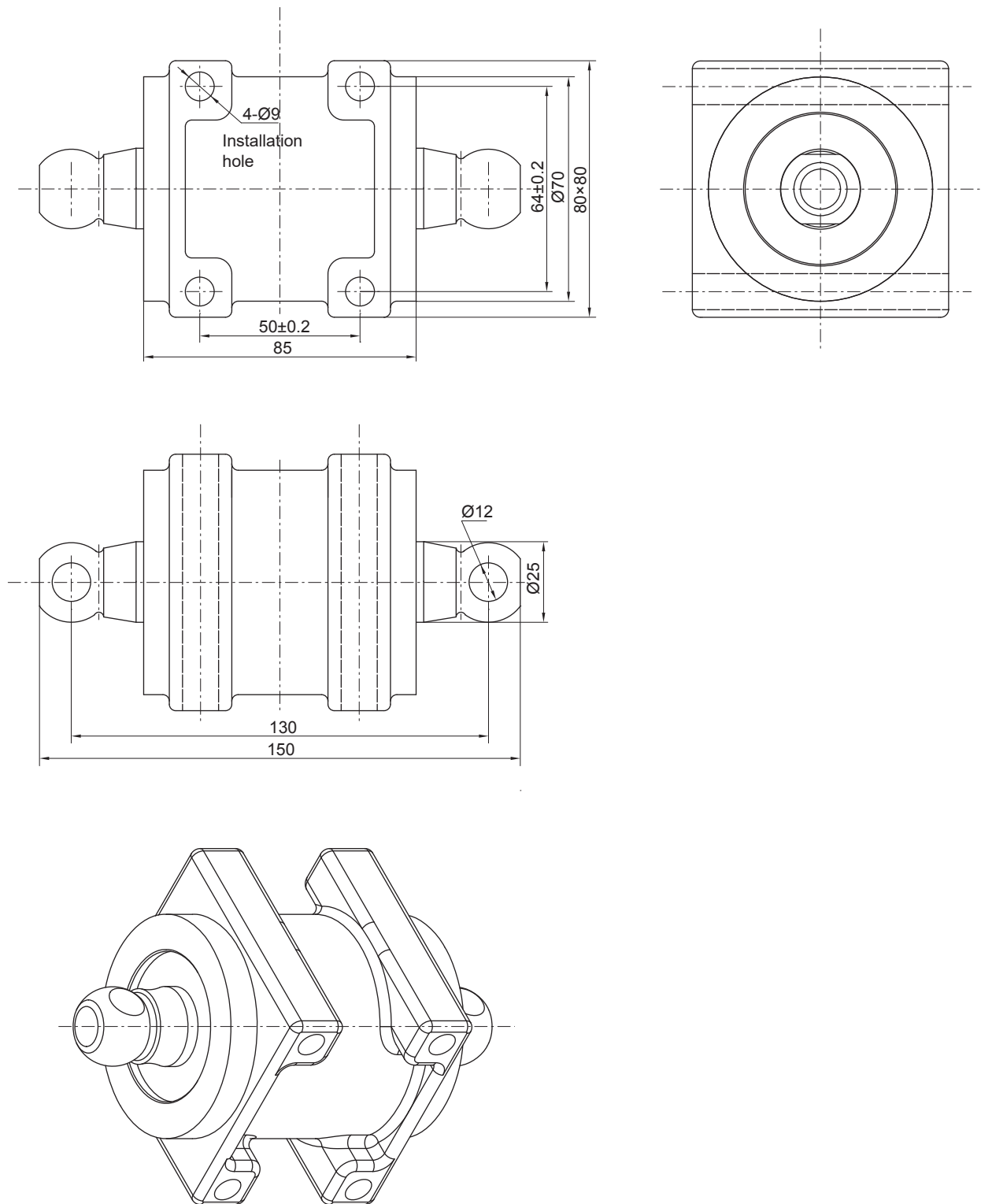
Unit: mm

Appendix 9 By-pass pipe structure



Unit: mm

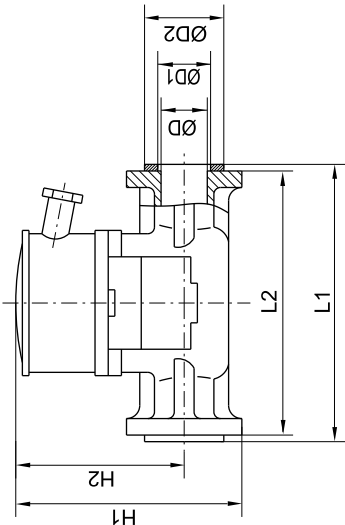
Appendix 10 Dimension of middle support box



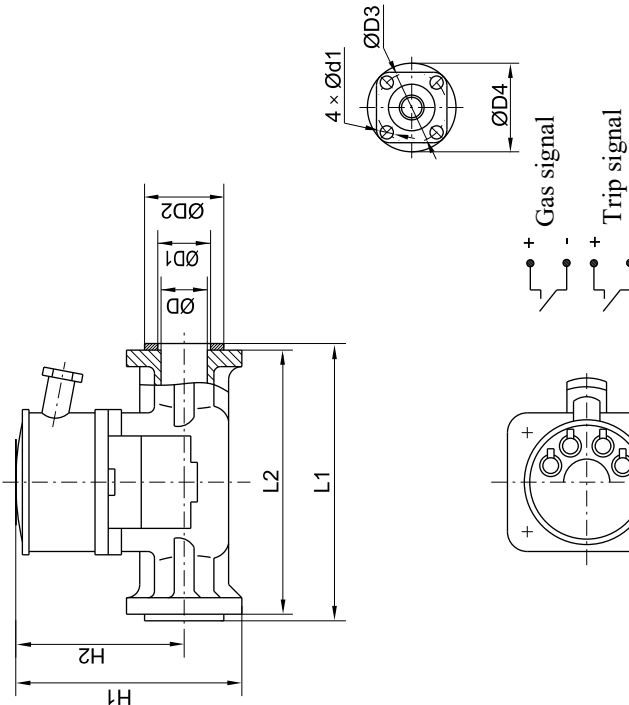
Unit: mm

Appendix 1 1 Dimension and installation of protective relay

QJ4-25A Protective relay



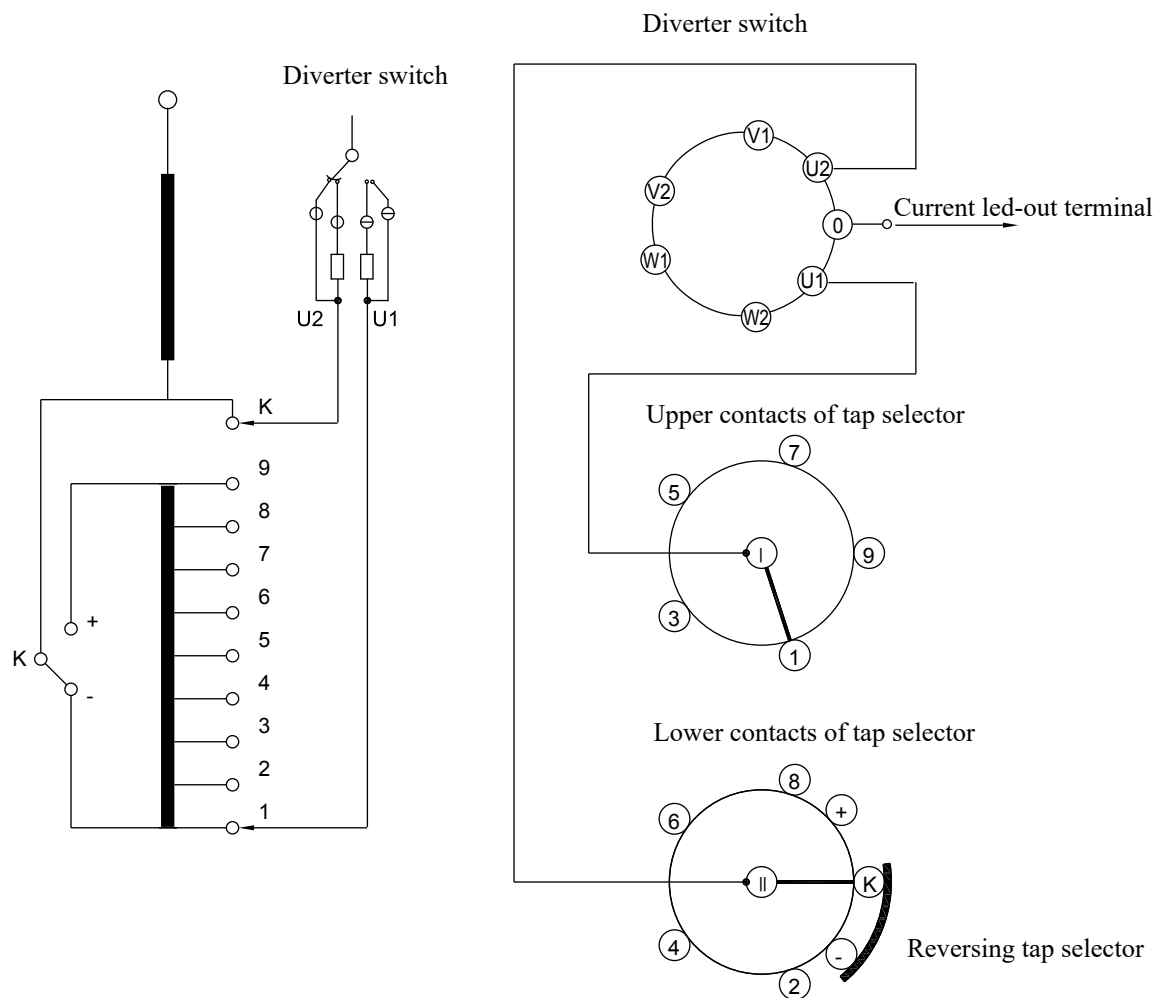
QJ4-25 Protective relay



Type	D	D1	D2	D3	D4	d1	H1	H2	L1	L2	Remarks
QJ4-25A	25	35	65	85	115	14	215	153	208	200	1 pair of gas signal and 1 pair of trip signal with gas collector, for OLTC
QJ4-25	25	35	65	85	115	14	215	153	208	200	1 pair of gas signal and 1 pair of trip signal, for OLTC

Unit: mm

Appendix 12 CM-III (10193W) operating position table and connection diagram



Operating position no.	19
Voltage no.	17
Set position	● 9b

Reversing tap selector	← K+ K- →																		
Tap changer	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
Tap selector contact	1	2	3	4	5	6	7	8	9	K	1	2	3	4	5	6	7	8	9
Indicating position	1	2	3	4	5	6	7	8	9a	9b	9c	10	11	12	13	14	15	16	17

Unit: mm



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